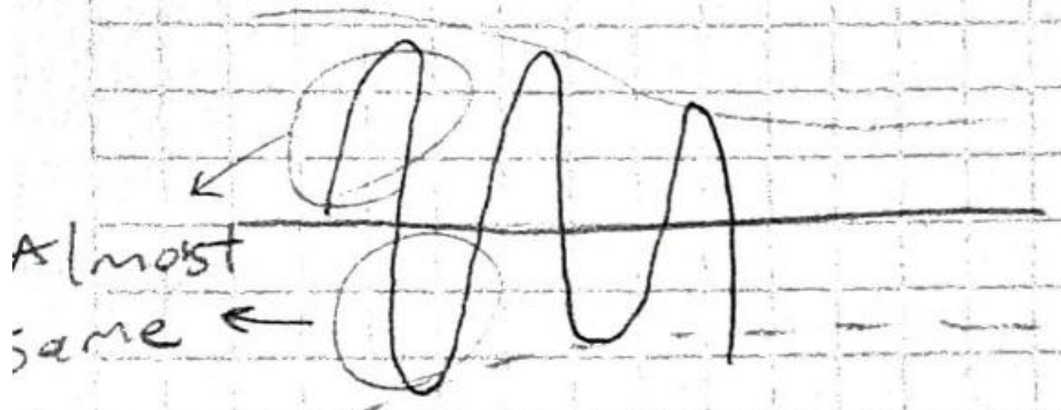
$$\lim_{w\to\infty} \int_{-\infty}^{\infty} f(t) \underbrace{\cos wt}_{\text{sin}wt}\, dt = 0$$

w is very big
(yani oscillates fast)

slow ↑ fast ↑

why zero?



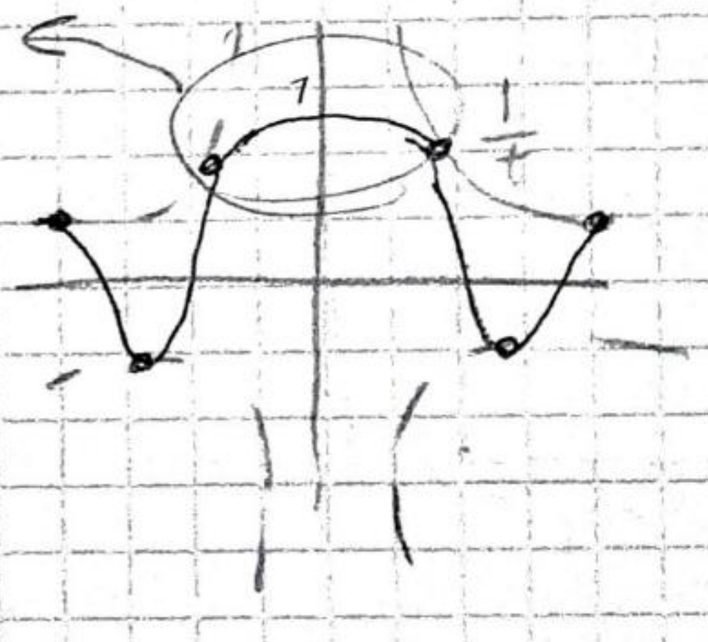
Sinc Function

$$Sinc(t) = \frac{\sin t}{t} = \frac{1}{t} \cdot \sin t$$

↓ envelope

→ oscillation in envelop

by L'hospital rule ←



$$\star \int_{-\infty}^{\infty} Sinc(t)\, dt = \pi$$

(very hard to integrate)

$$\star \int_{e}^{\infty} \frac{\sin t}{t}\, dt = \frac{\pi}{2}$$

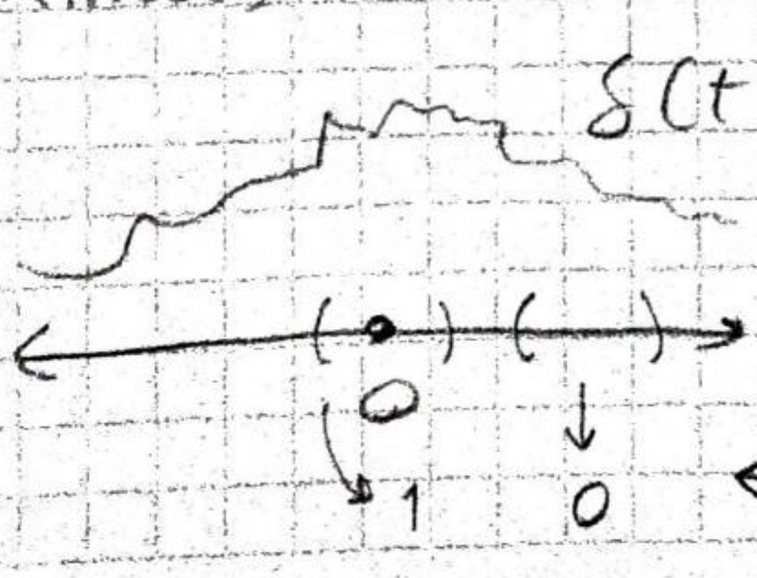
Dirac Deltas

$\delta(t)$ $\delta[n]$.

↑ ↑ Kronecker Delta

Dirac Delta (discrete)

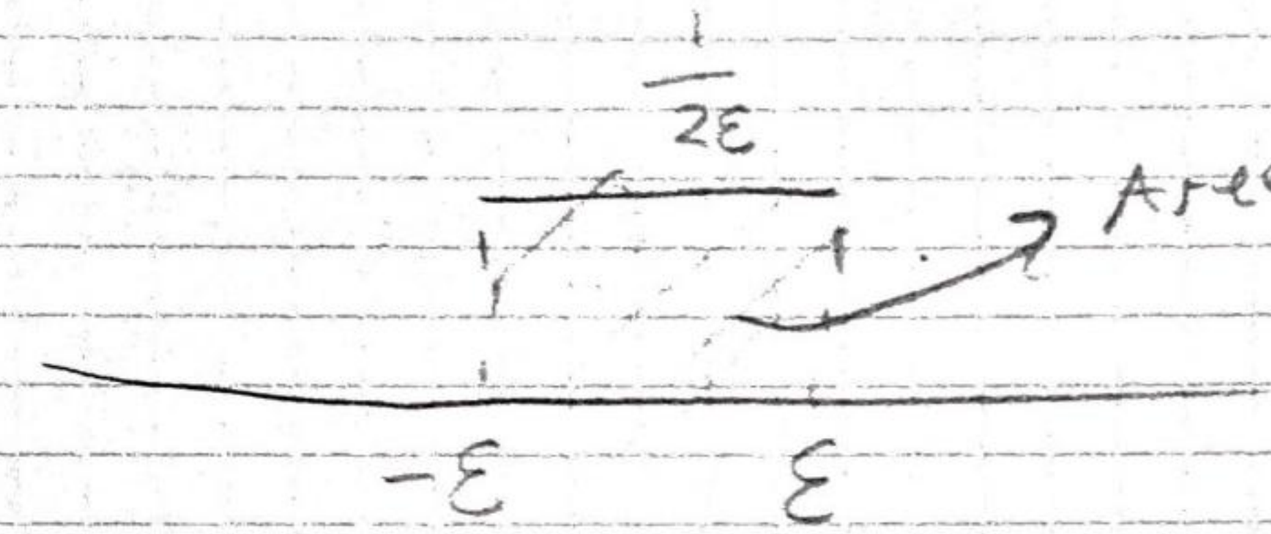
(continuous)



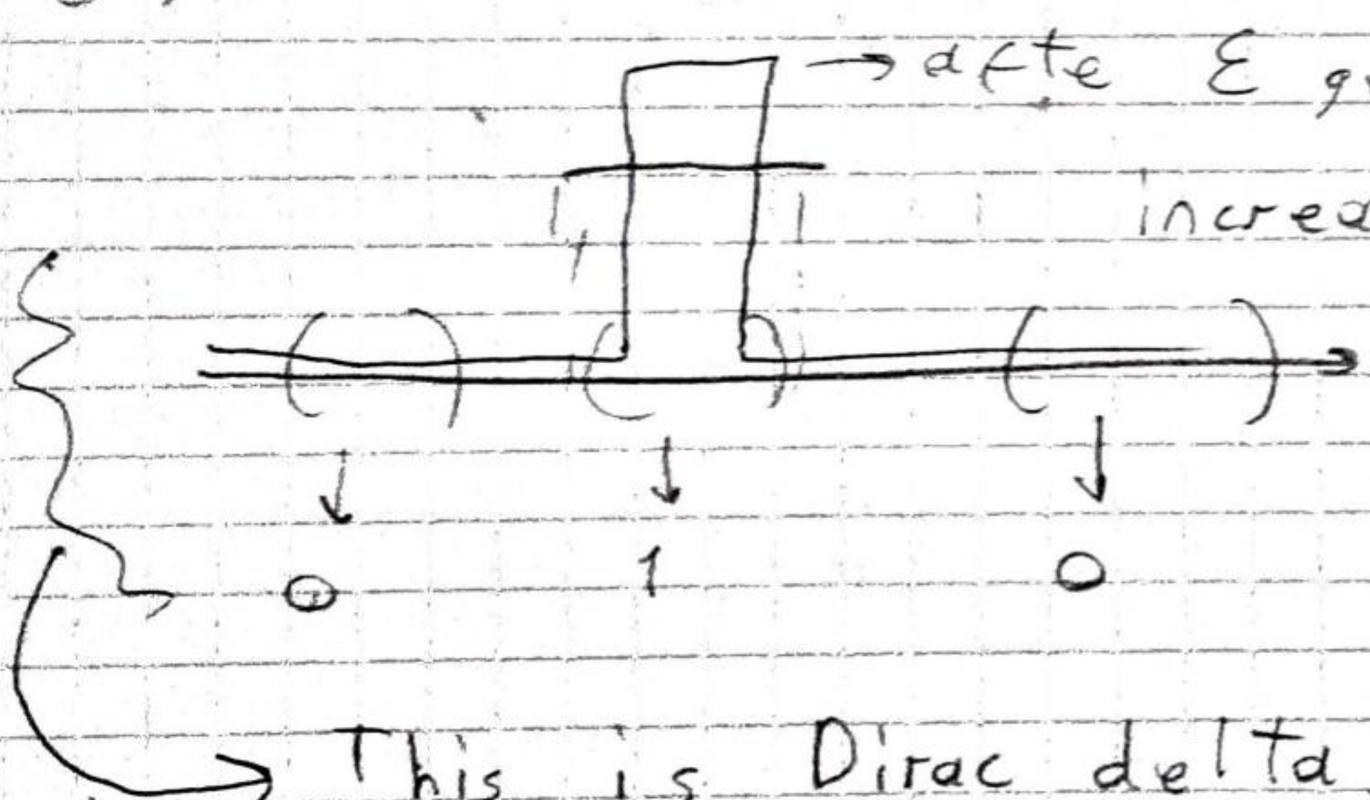
$$\int_{A}^{B} \delta(t)\, dt = \begin{cases} 0, & \text{if } 0 \notin (a,b) \\ 1, & \text{if } 0 \in (a,b) \end{cases}$$

örnek

$$\frac{1}{2\varepsilon}\chi_{(-1,1)}\left(\frac{t}{\varepsilon}\right) = \frac{1}{2\varepsilon}\chi_{(-\varepsilon,\varepsilon)}(t)$$



$$\lim_{\varepsilon \to 0} \frac{1}{2\varepsilon}\chi_{(-\varepsilon,\varepsilon)}(t)$$



→ after ε gets smaller → height increase. Area is still 1

→ This is Dirac delta!

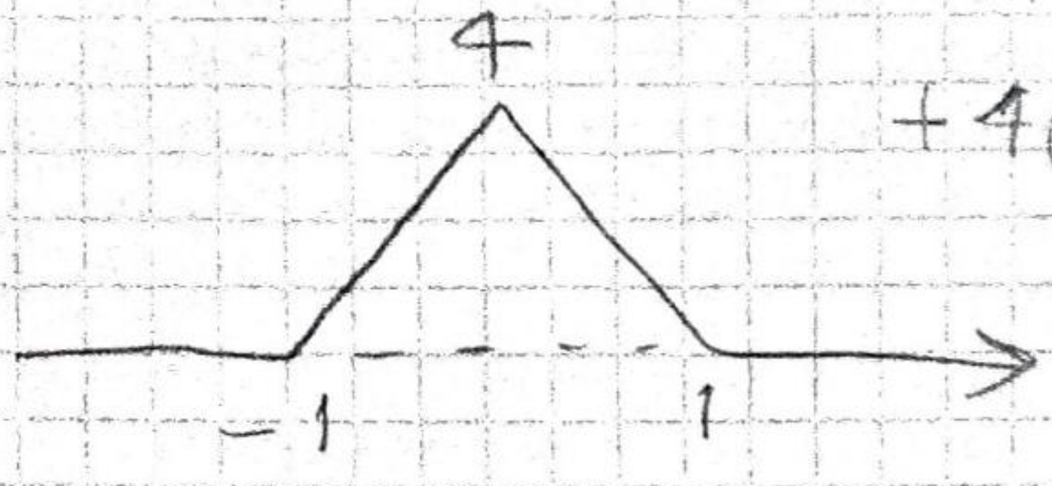
$$\delta(t) = \lim_{\varepsilon \to 0} \frac{1}{A\varepsilon} f\left(\frac{t}{\varepsilon}\right)$$

$$\frac{1}{1+t^2} \rightarrow \delta(t) = \lim_{\varepsilon \to 0} \frac{1}{A} \frac{1}{\varepsilon} \frac{1}{1+\left(\frac{t}{\varepsilon}\right)^2}$$

Example

$4(t+1)\chi_{(-1,0)}(t)$

$+4(1-t)\chi_{(0,1)}(t)$



Convert this to Dirac Delta!

$\delta(t) =$

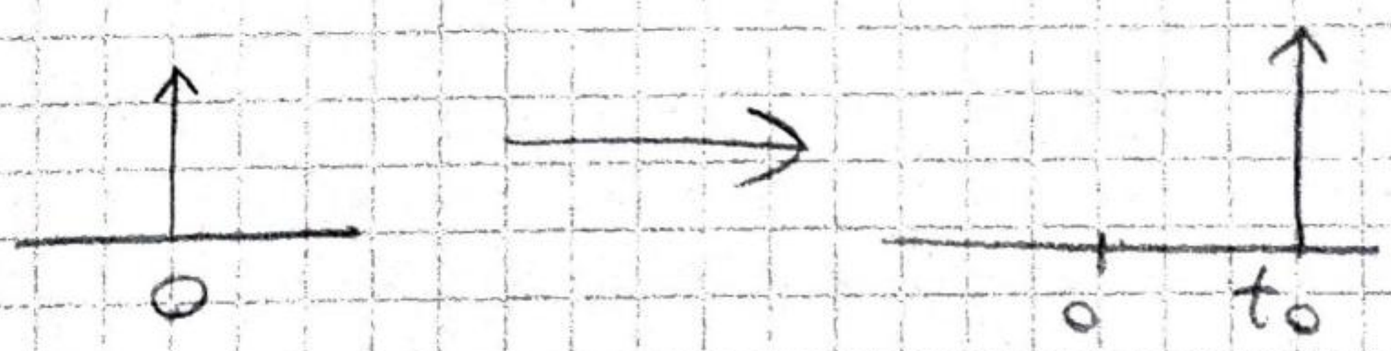
$\lim_{\varepsilon\to 0} \frac{1}{4\varepsilon}\left[4\left(\frac{t}{\varepsilon}+1\right)\chi_{(-1,0)}\left(\frac{t}{\varepsilon}\right) + 4\left(1-\frac{t}{\varepsilon}\right)\chi_{(0,1)}\left(\frac{t}{\varepsilon}\right)\right]$ ✓

↳ don't forget this part

$\chi_{(-\varepsilon,0)}(t)$ $\qquad$ $\chi_{(0,\varepsilon)}(t)$

Propetries of Dirac Delta

1) $\delta(t) \longrightarrow \delta(t-t_0)$



2) $\delta\left(\frac{t}{a}\right) = \frac{1}{|a|}\delta(t)$

3) $f(t)\cdot\delta(t) = f(0)\cdot\delta(t)$

$f(t)\cdot\delta(t-t_0) = f(t_0)\cdot\delta(t-t_0)$

4) $\int_{-\infty}^{\infty} f(t)\cdot\delta(t-t_0)\,dt = \int_{-\infty}^{\infty} f(t_0)\,\delta(t-t_0)\,dt$

$f(t_0)$ by take it out!

$= f(t_0)\int_{-\infty}^{\infty}\delta(t-t_0)dt = f(t_0)$

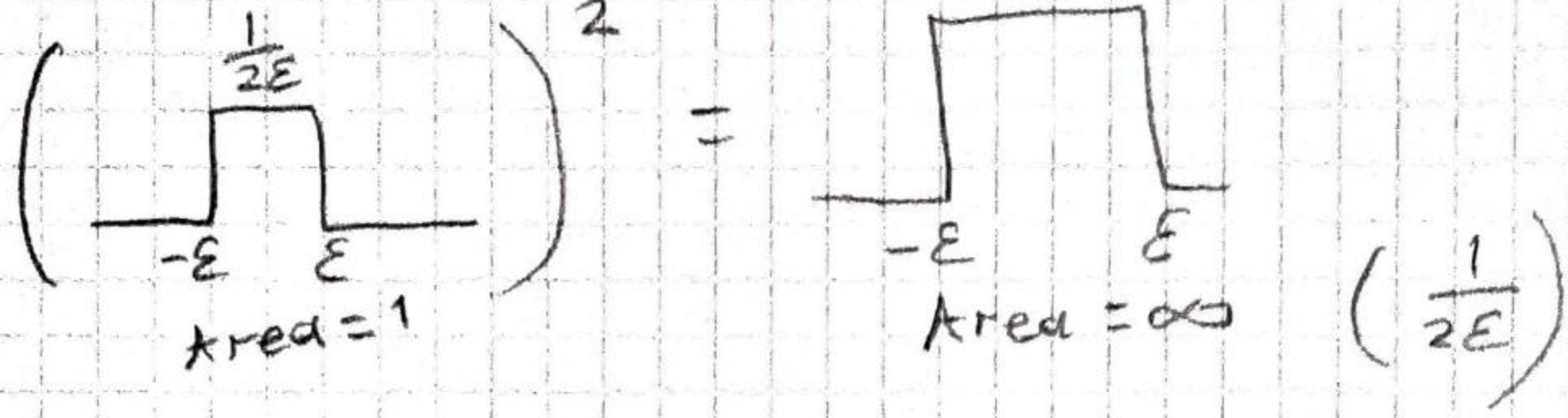
1

★ Summary: Dirac Delta kills the integral.

Properties of $\delta(t)$

1- Dirac Delta is not a function. It's a distribution.

2- Square of impulse (Dirac delta) is not defined

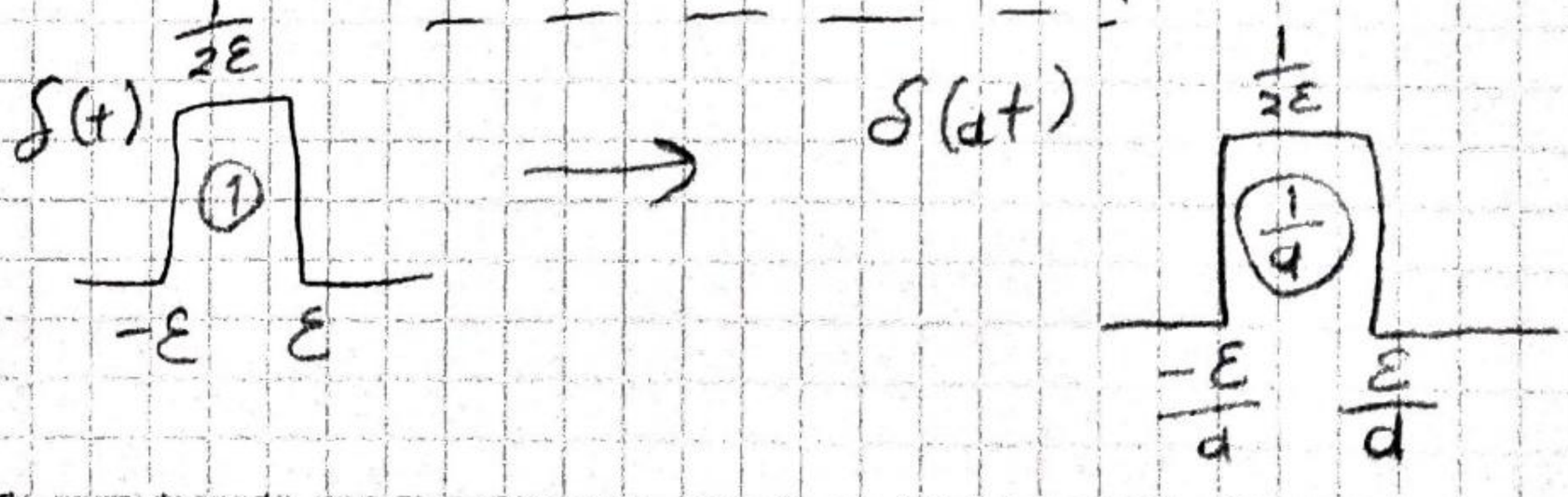


3- Multiplying with a constant

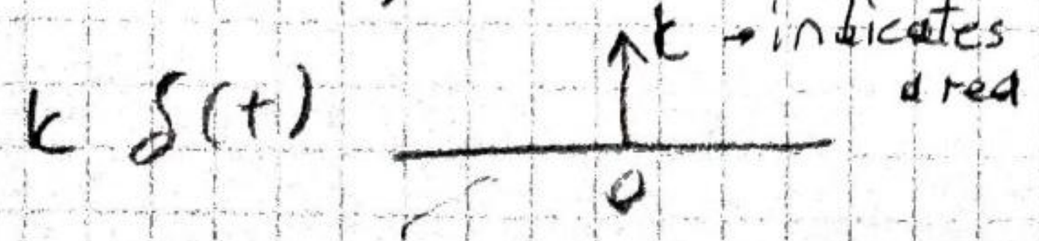
$-3\,\delta(t) \rightarrow$ area -3

4- Shifting

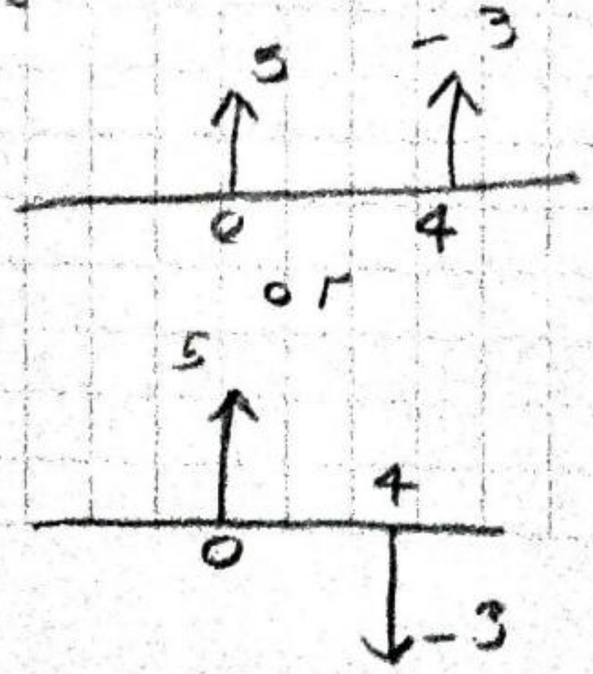
5- Scaling $\left| \quad f(at) = \frac{1}{|a|}\delta(t) \right.$



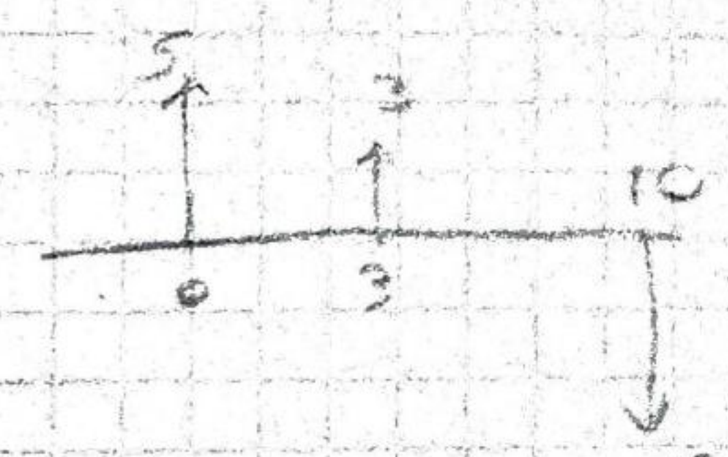
6- Plotting dirac delta:



$5\delta(t) - 3\delta(t-4)$



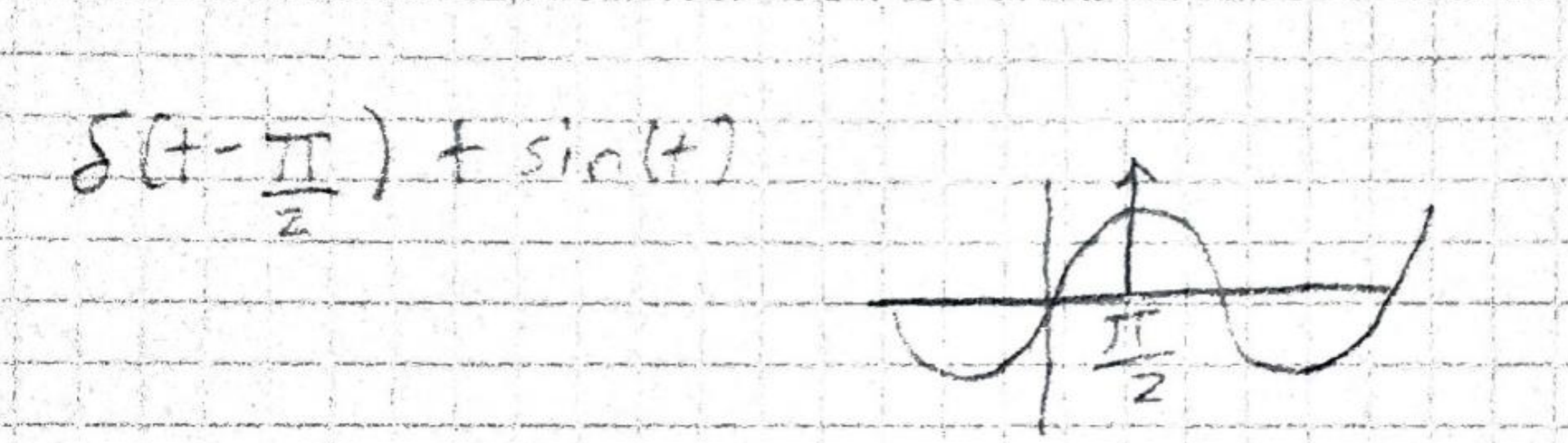
$5\,\delta(t) + 3\,\delta(t-3) - 6\,\delta(t-10)$

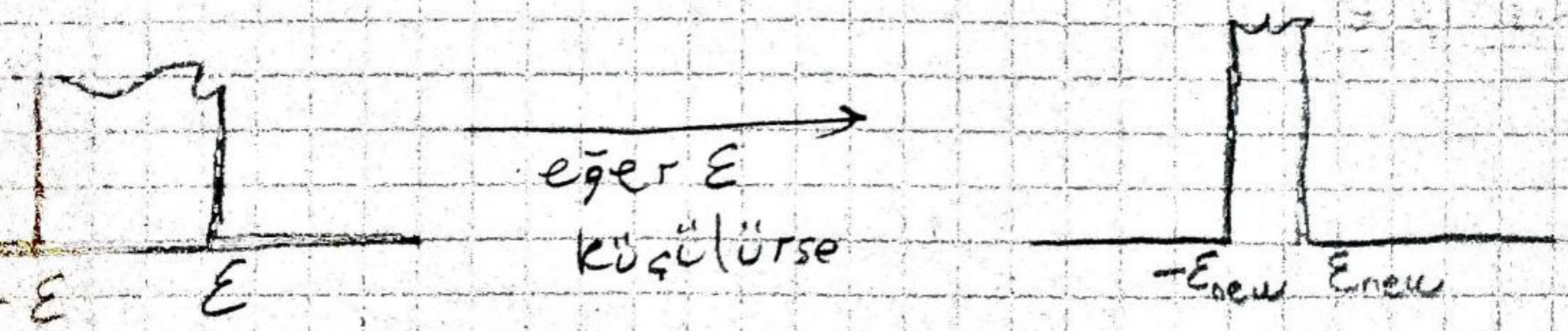


$\delta(t) + \delta(t-4) + u(t)$

$\delta\left(t-\frac{\pi}{2}\right) + \sin(t)$

$x(t)\,\delta(t) = x(0)\,\delta(t)$





8- Dirac delta destroys integrals (it's a friend).

$$\int_{-\infty}^{\infty} \sqrt{\frac{e^{-t^2}\sin(t)}{t^4+10}}\,\delta(t-5)\,dt = \frac{e^{-5^2}\sin(5)}{5^4+10}$$

Rule! (madde 7'deki kural)

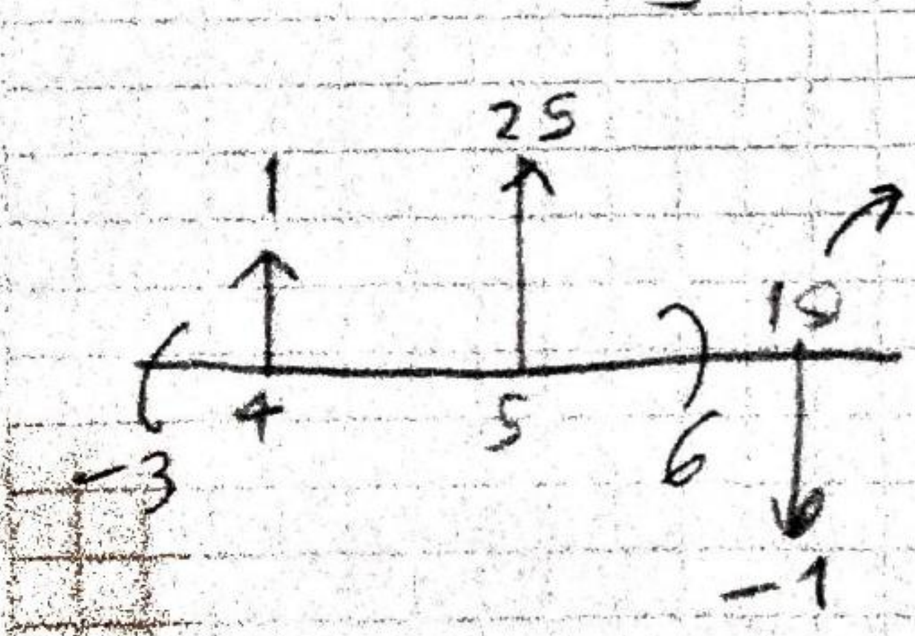
$$\int_A^B f(t)\,\delta(t-a)\,dt = \int_A^B f(a)\,\delta(t-a)\,dt = f(a)\int_A^B \delta(t-a)\,dt$$

$$= \begin{cases} f(a) & \text{if } a \in [A,B] \\ 0 & \text{if } a \notin [A,B] \end{cases}$$

0 if $a \notin [A,B]$

1 if $a \in [A,B]$

example $\int_{-3}^{6} [\delta(t-4) + t^2\,\delta(t-5) - \delta(t-10)]\,dt$



doesn't count!

$= 26$

Derivative of Unit Step

$$\frac{du(t)}{dt}\Big|_{t=0} = ?$$

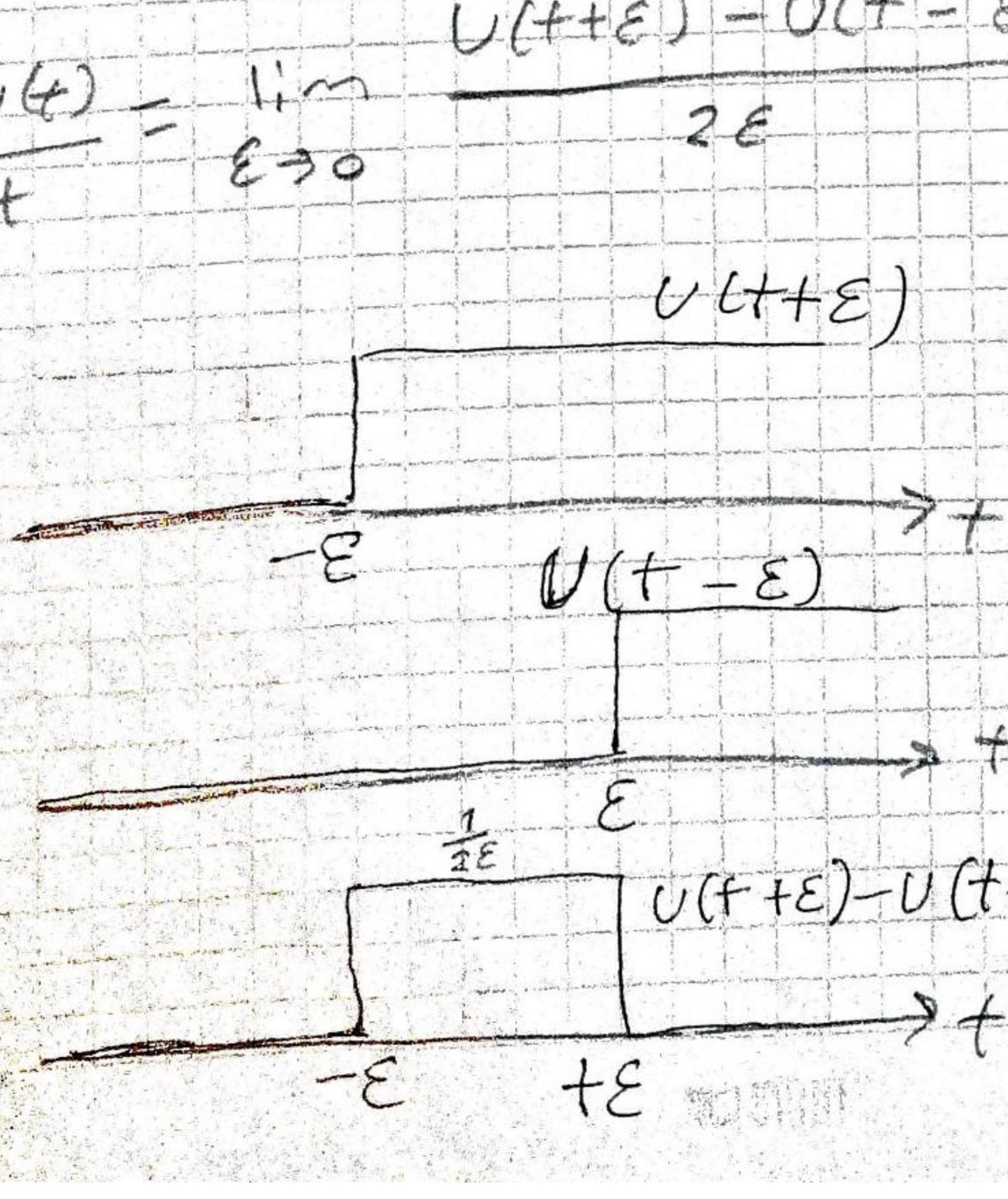
$$= \left| \frac{du(t)}{dt} = \delta(t) \right|$$

Rule! Derivative of discontinuities give dirac deltas!

$$\star \quad \frac{df(t)}{dt} = \lim_{\varepsilon \to 0} \frac{f(t+\varepsilon) - f(t-\varepsilon)}{2\varepsilon}$$

for dirac delta:

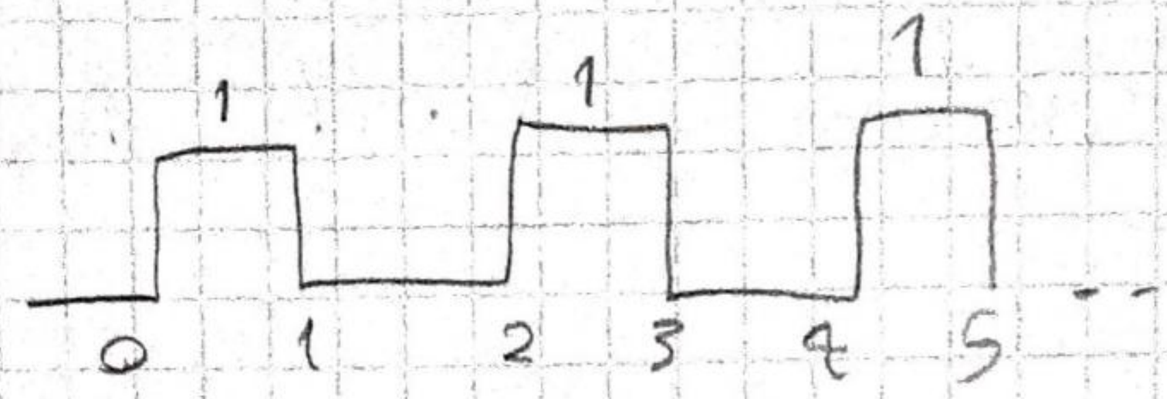
$$\frac{du(t)}{dt} = \lim_{\varepsilon \to 0} \frac{u(t+\varepsilon) - u(t-\varepsilon)}{2\varepsilon}$$



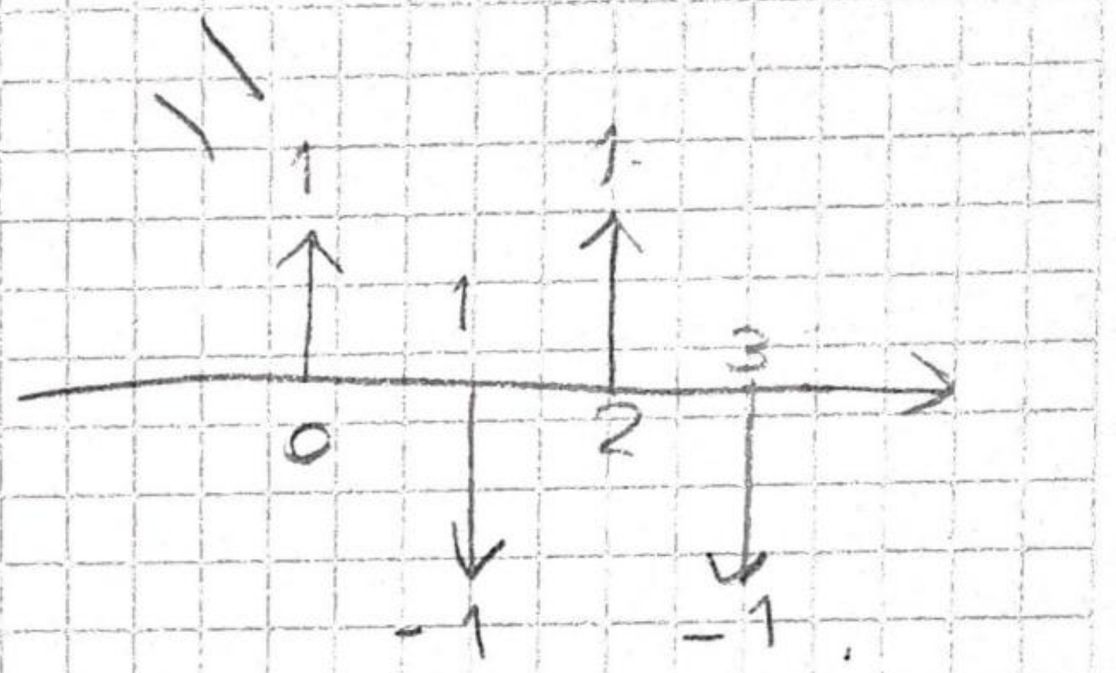
$$\lim_{\varepsilon \to 0} \frac{u(t+\varepsilon) - u(t-\varepsilon)}{2\varepsilon}$$

$$= \lim \frac{\chi_{(-\varepsilon,\varepsilon)}(t)}{2\varepsilon} = \delta(t)$$

example



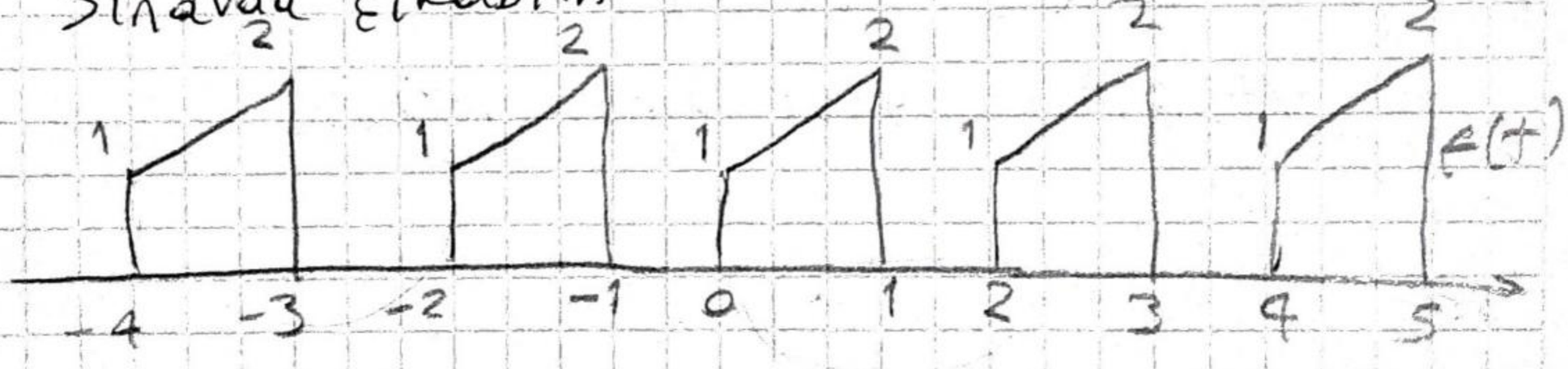
$$\frac{d}{dt}\left(\sum_{k=-\infty}^{\infty} \chi_{(0,1)}(t-2k)\right) = ?$$



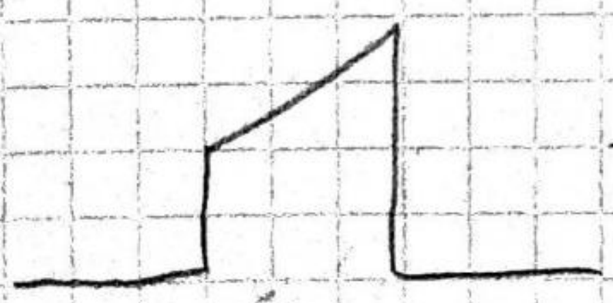
$$= \sum_{k=-\infty}^{\infty} (-1)\,\delta(t-k)$$

★★★★★ Sınavda çıkabilir

example



a) f(t) ? (analytically

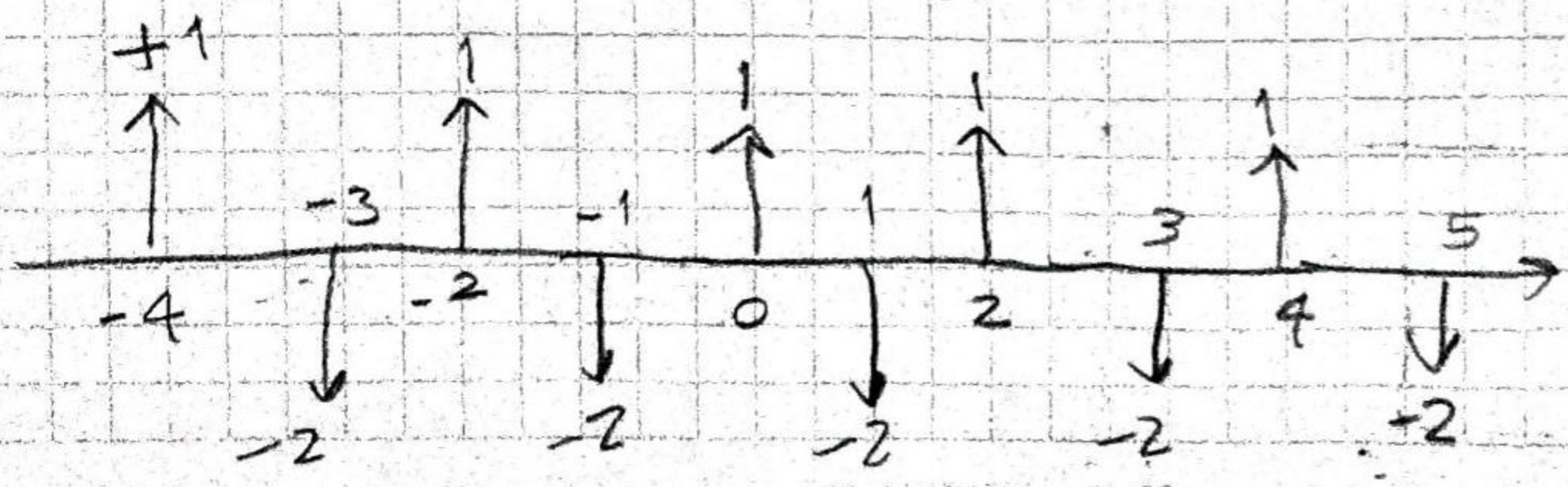


$\rightarrow (t+1)\,\chi_{(0,1)}(t$

$$\text{general} = \sum_{k=-\infty}^{\infty} (t-2k+1)\,\chi_{(0,1)}(t-2k)$$

b) find $\frac{df(t)}{dt}$ and plot it.

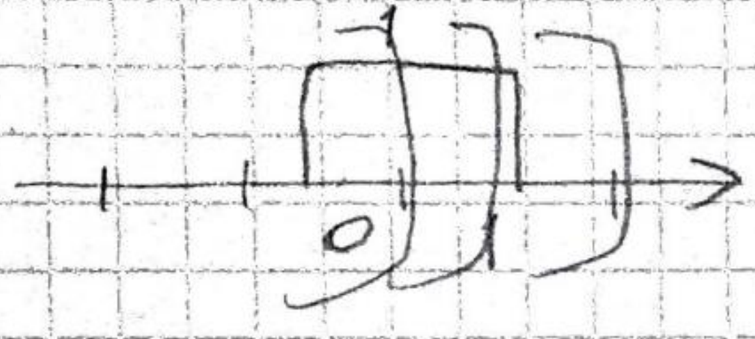
(look for derivatives on points)



Integration

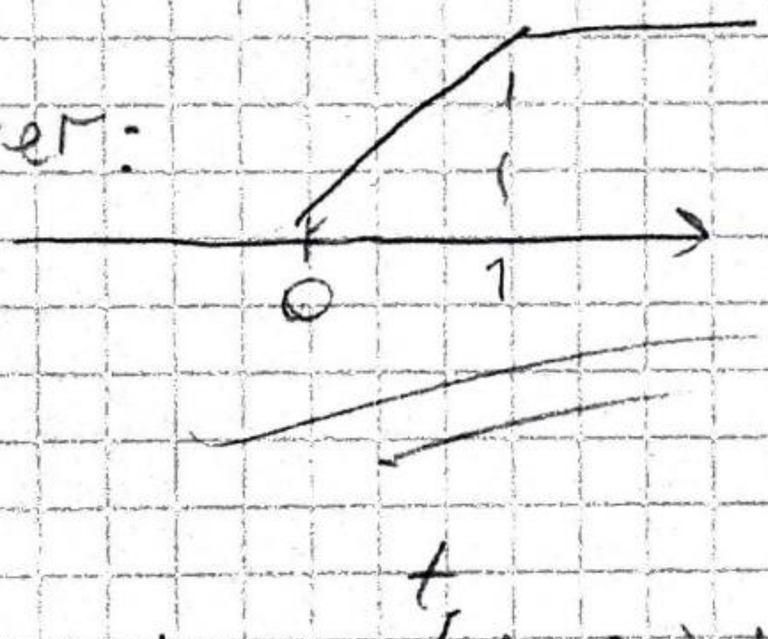
example

Find the integral of $\chi_{(0,1)}(t)dt$

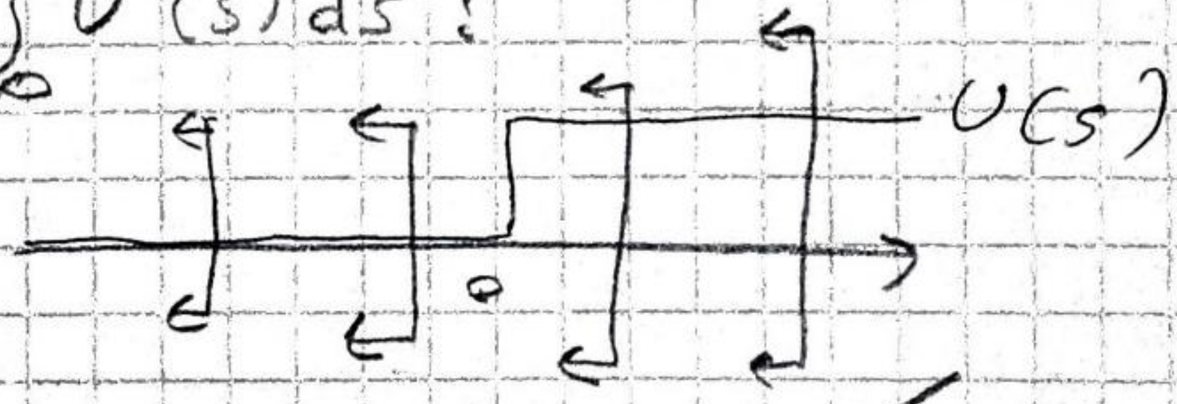


$$\int_{-\infty}^{t} \chi_{(0,1)}(s)\,ds$$

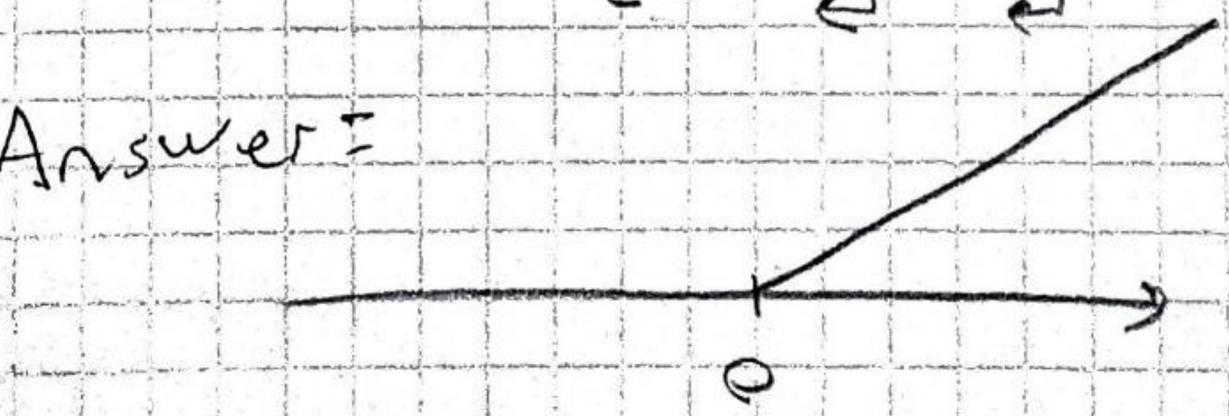
Answer:



example $\int_{-\infty}^{t} U(s)\,ds$?

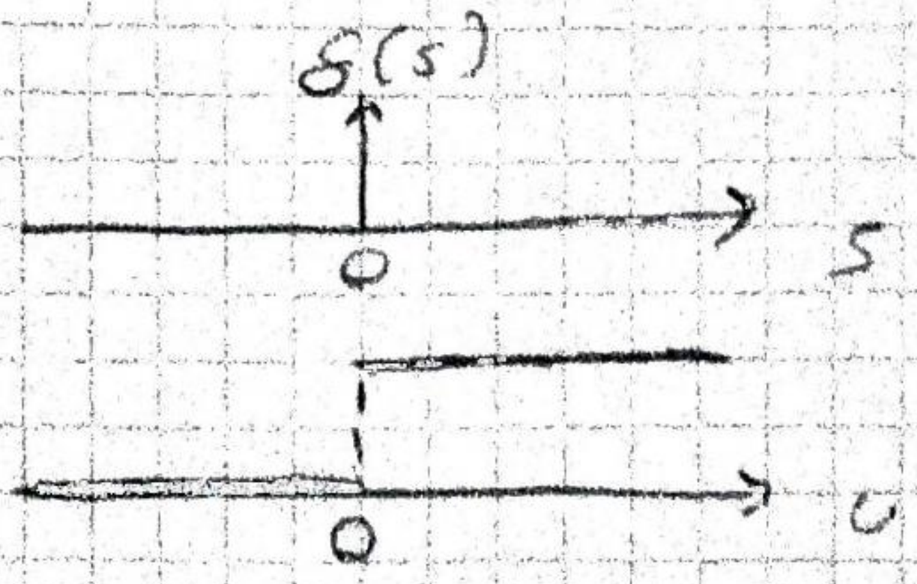


Answer:

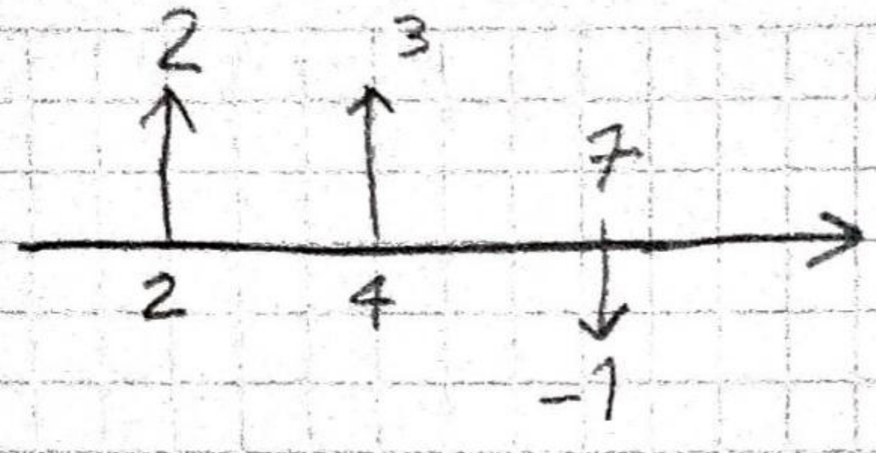


★ Claim: Integral of dirac delta is unit step.

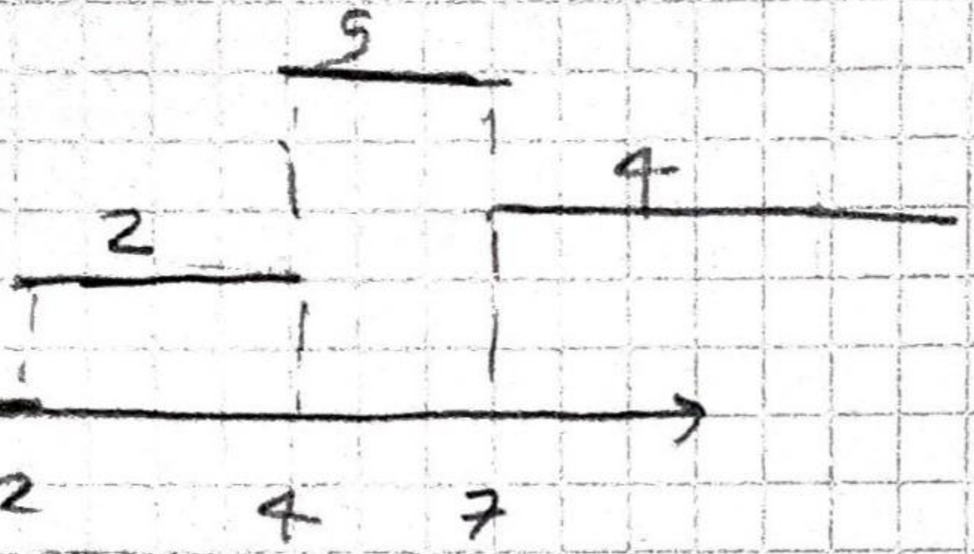
$$\int_{-\infty}^{t} \delta(s)\,ds = U(t)$$



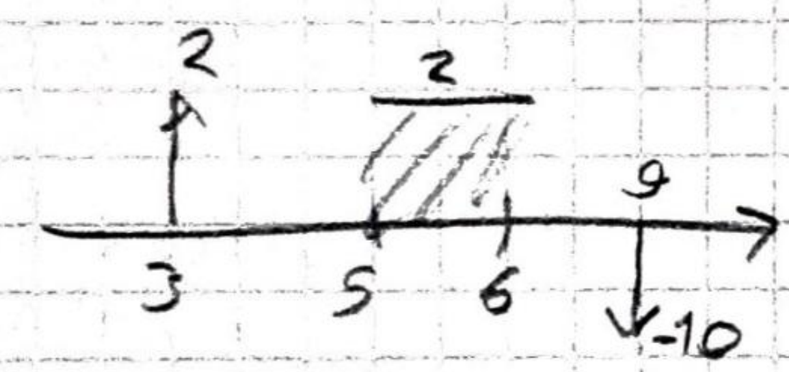
ex

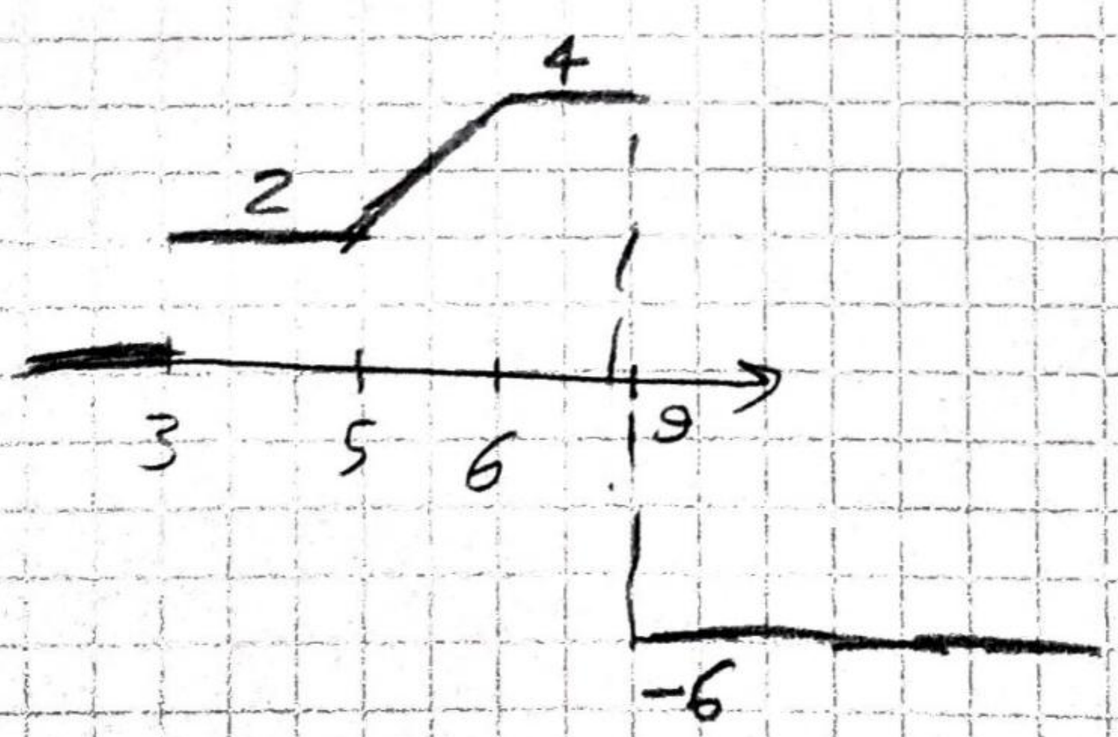


integrated to



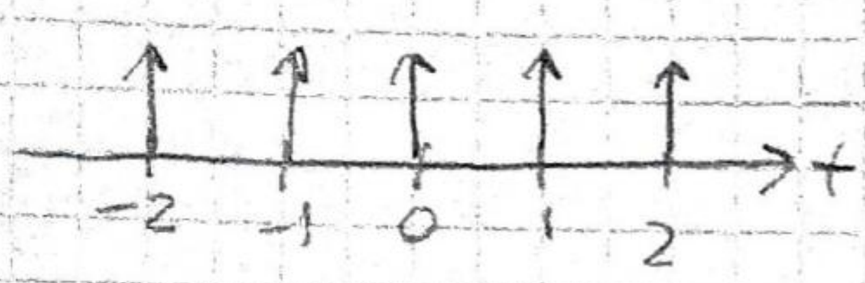
example





Periodic Dirac Delta

Dirac delta train:



$$\cdots + \delta(t+2) + \delta(t+1) + \delta(t) + \delta(t-1) + \cdots$$

$$\sum_{n=-\infty}^{\infty} \delta(t-n) \quad \text{or} \quad \sum_{n=-\infty}^{\infty} \delta(t+n)$$

★ Geometric Series

$$S = 1 + r + r^2 + r^3 + \cdots$$

$$\rightarrow \quad rS = r + r^2 + r^3 + r^4 + \cdots$$

$$S - rS = 1$$

$$S = \frac{1}{1-r}$$

★ $|r| < 1$

★ Devami:

$$S = 1 + r + r^2 + \cdots + r^N$$

$$rS = r + r^2 + r^3 + \cdots + r^{N+1}$$

$$S(1-r) = 1 - r^{N+1}$$

$$S = \frac{1 - r^{N+1}}{1-r}$$

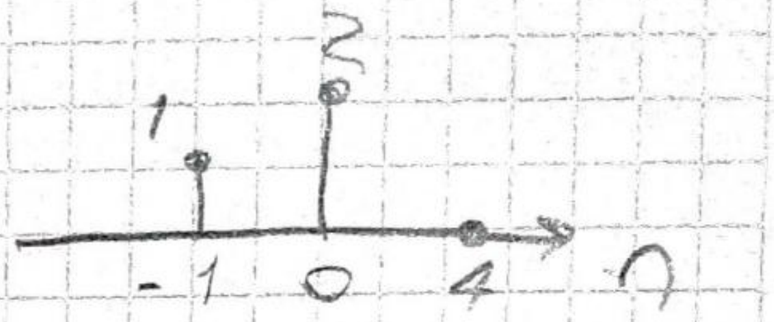
★ no limits on r

★ $\sum_{k=-\infty}^{\infty} e^{2\pi i k t} = \sum_{n=-\infty}^{\infty} \delta(t-n)$

Discrete Signals

$x(t)$ — continuous

$x[n]$ — discrete

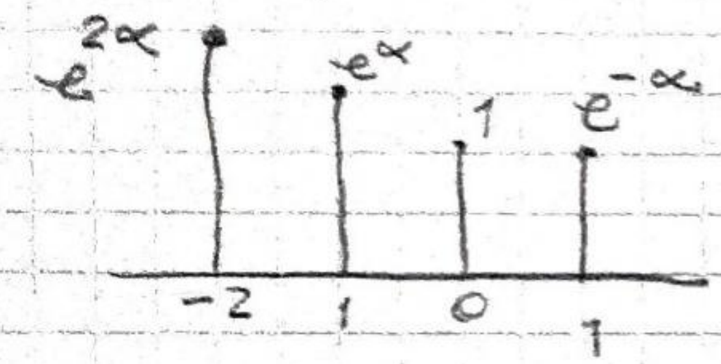


$(\underline{1}\ 2\ 0\ 0)$

↳ zeroth point is underlined

Discrete exponentials

$x[n] = e^{-\alpha n} \quad \alpha > 0$

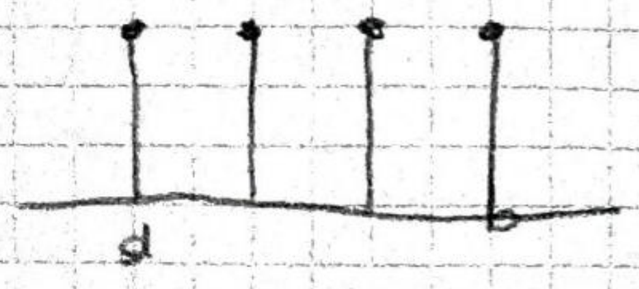


Discrete Sinusoids

$\sin(\omega t) \qquad \sin[\Omega n]$

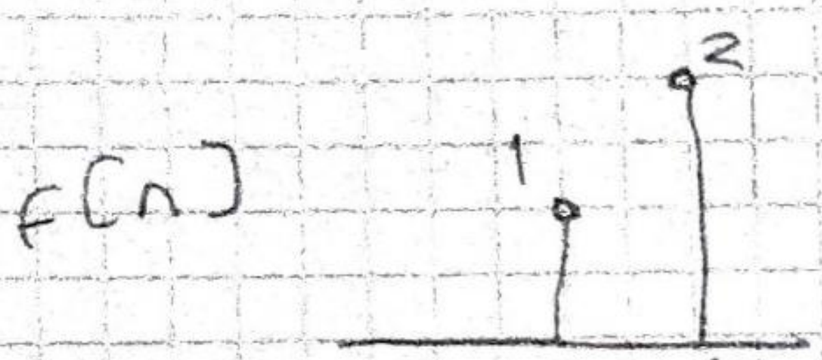
indicator

$$\chi_{(a,b)}[n] = \begin{cases} 1 & \text{if } a \le n \le b \\ 0 & \text{elsewhere} \end{cases}$$



Shifts & Scales

scales Not possible in discrete functions!



f[2n] ?

for 1

$n = \frac{1}{2}$

not doable!

* Shifting → only integer shifts are possible

Discrete Dirac Delta

$$\delta[n] \begin{cases} 1 & n=0 \\ 0 & \text{elsewhere} \end{cases}$$

Discrete dirac delta = kronecker delta

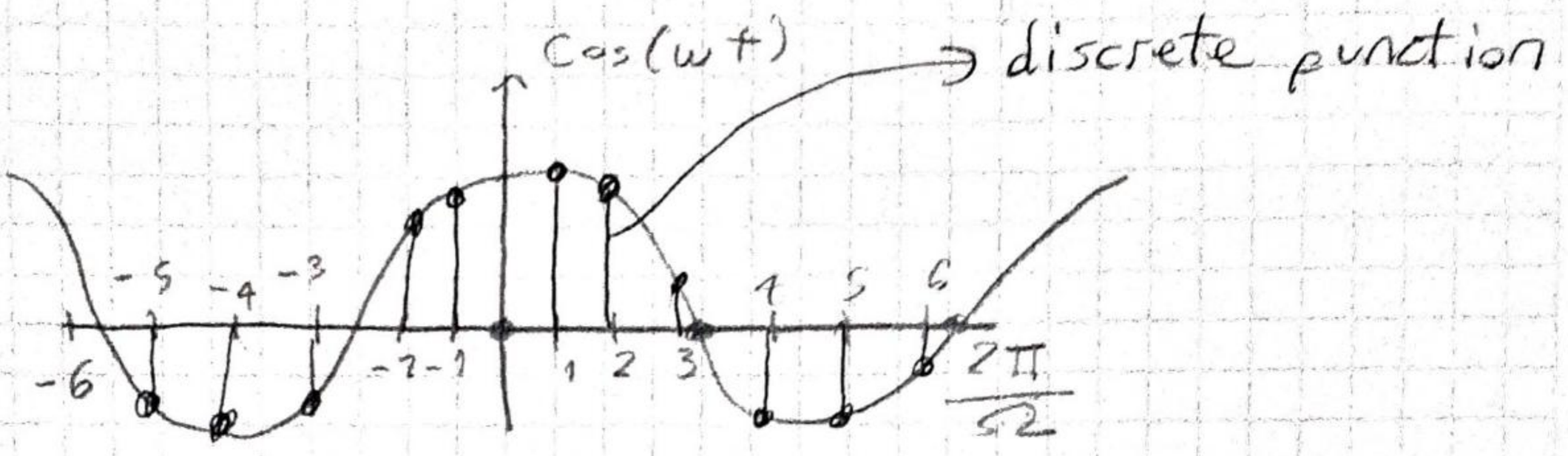
Sinc

$$\text{Sinc}[n] = \frac{\sin[n]}{n}$$

Periodicity of Complex Exponentials (Sinusoids)

$\cos(\omega t)$	$\cos[\Omega n]$
1. Periodic for all ω.	1. Can be non-periodic for some Ω.
2. $T = \frac{2\pi}{\omega}$	2. $T \neq \frac{2\pi}{\Omega}$ in most cases.

How to get $\cos[\Omega n]$



Definition of periodicity of $f(t)$

$$f(t) = f(t+T) \quad \forall n$$

$$\cos[\Omega n] = \cos[\Omega (n+N)]$$ → must be an integer!

my function will be periodic if I can find such an N.

$$\boxed{\Omega = \frac{2k\pi}{N}}$$

ex

$\sin\left[\frac{6\pi}{31} n\right]$ ✓ $(T=31)$ $\sin[\sqrt{3}\, n]$ X not periodic

$\sin[5n]$ X no π

Continuous sinusoids	Discrete sinusoids
1- $\cos(\omega t)$	1- $\cos[\Omega n]$
2- Always periodic	2- Not always periodic
3- Periodic is given by $T = \frac{2\pi}{\omega}$	3- Period is NOT given by $T = \frac{2\pi}{\Omega}$
4- $\cos((\omega+\Delta)t)$ is never equal to $\cos(\omega t)$ unless $\Delta = 0$.	4- I can find L such that $\cos[\Omega n] = \cos[\Omega+L]n$

*) $\cos(\Omega n)$ periodic?

$e^{i\Omega n} = e^{i\Omega(n+N)}$ ← integer

→ for periodicy, I must find (be able to find) such an N.

if $N=1$ the discrete func. is constant. → $x[n+1] = x[n]$

$N=2$ period for smallest possible non constant discrete func.

$e^{i\Omega n} = e^{i\Omega n} \cdot e^{i\Omega N} \rightarrow e^{i\Omega N} = 1$

$\Omega N = 2k\pi$

↳ integer

$\boxed{\Omega = \frac{2k\pi}{N}}$ see the previous page (recap)

$\star\ e^{i(\Omega+\Delta)n} = e^{i\Omega n}$ → $\cos[(\Omega+2k\pi)n] = \cos[\Omega n]$

$e^{i\Omega n} \cdot e^{i\Delta n} = e^{i\Omega n}$

$e^{i\Delta n} = 1$ for all n

$\Delta = 2k\pi$

example

$\cos\left[\frac{7\pi}{2}n\right] \stackrel{?}{=} \cos\left[\frac{3\pi}{2}n\right]$ ✓ (because n is integer)

$\stackrel{?}{=} \cos\left[\frac{15\pi}{2}n\right]$ ✓

example

Find the period of $\cos\left[\frac{3\pi}{7}\right]$. remember $\Omega = \frac{2k\pi}{N}$

$\frac{2k\pi}{N} \rightarrow \frac{2.3\pi}{2.7}$ → period. (N)

example

$\cos\left[\frac{17\pi n}{4}\right]$ $\cos\left[\frac{17\pi n}{4}\right] = \cos\left[\frac{2.17\pi n}{8}\right]$

period

simplification way:

$\cos\left[\frac{17\pi n}{4}\right] = \cos\left[\frac{\pi n}{4}\right] = \cos\left[\frac{2\pi n}{8}\right]$

(2π'in katlarını çıkardık)

Systems

signal

i = 9 →

```
scanf("%d\n", &i)
for i=1 to 10 {
    printf("%d\n", i);
    i++;
}
```

→ output

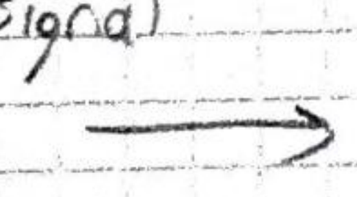
1
2
3
4
.
.
.

input signal

x(t)

x[t]

→ System



Economy machine program

→ output signal

y(t)

y[t]

We will be interested with linear and time invariant systems → LTI

If the system is LTI, we have a melted to estimate it without knowing its interval details.

- An analogy to signals and systems from linear algebra

LTI systems ↔ matrices

A, B, C

x(t) ↔ vectors

x[n] V_1, V_2, V_3

Linearity

A system is linear iff

$x_1(t) \rightarrow \boxed{\quad} \rightarrow y_1(t)$

$x_2(t) \rightarrow \boxed{\quad} \rightarrow y_2(t)$

$a x_1(t) + b x_2(t) \rightarrow \boxed{\quad} \rightarrow a y_1(t) + b y_2(t)$

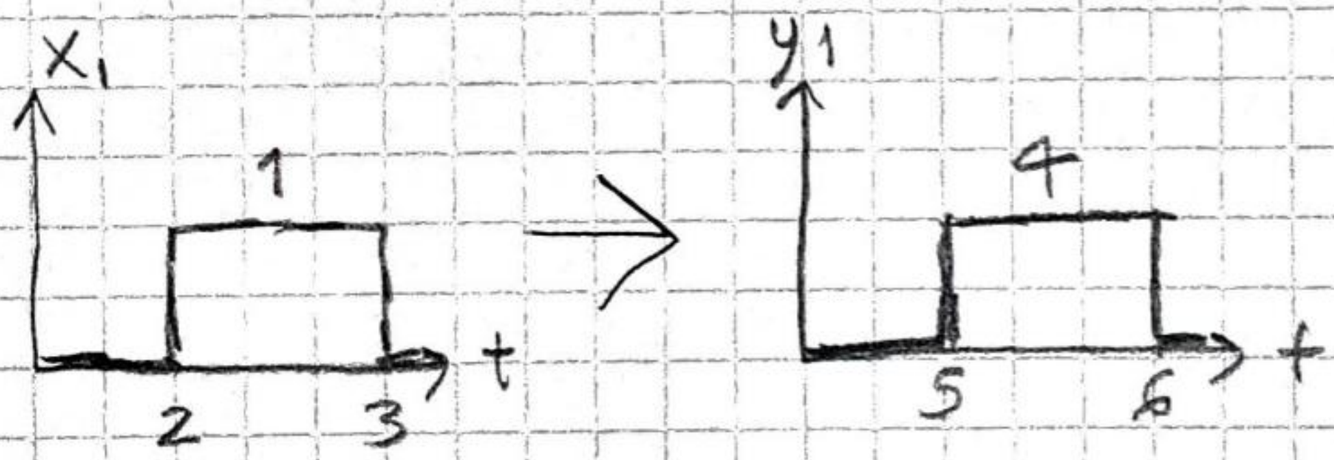
for every $a, b \in R$

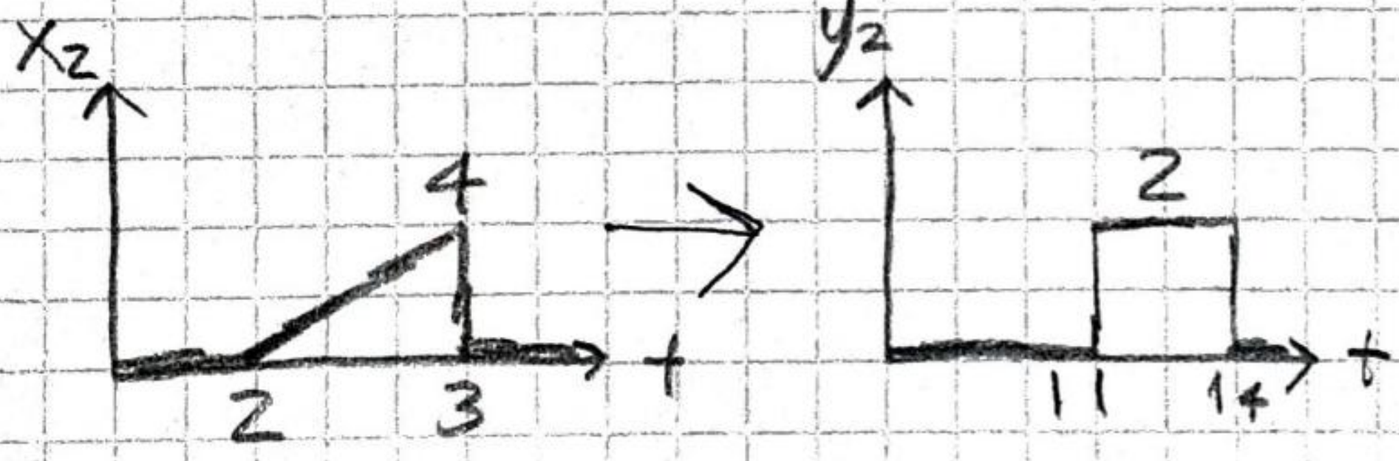
Matrix analogy:

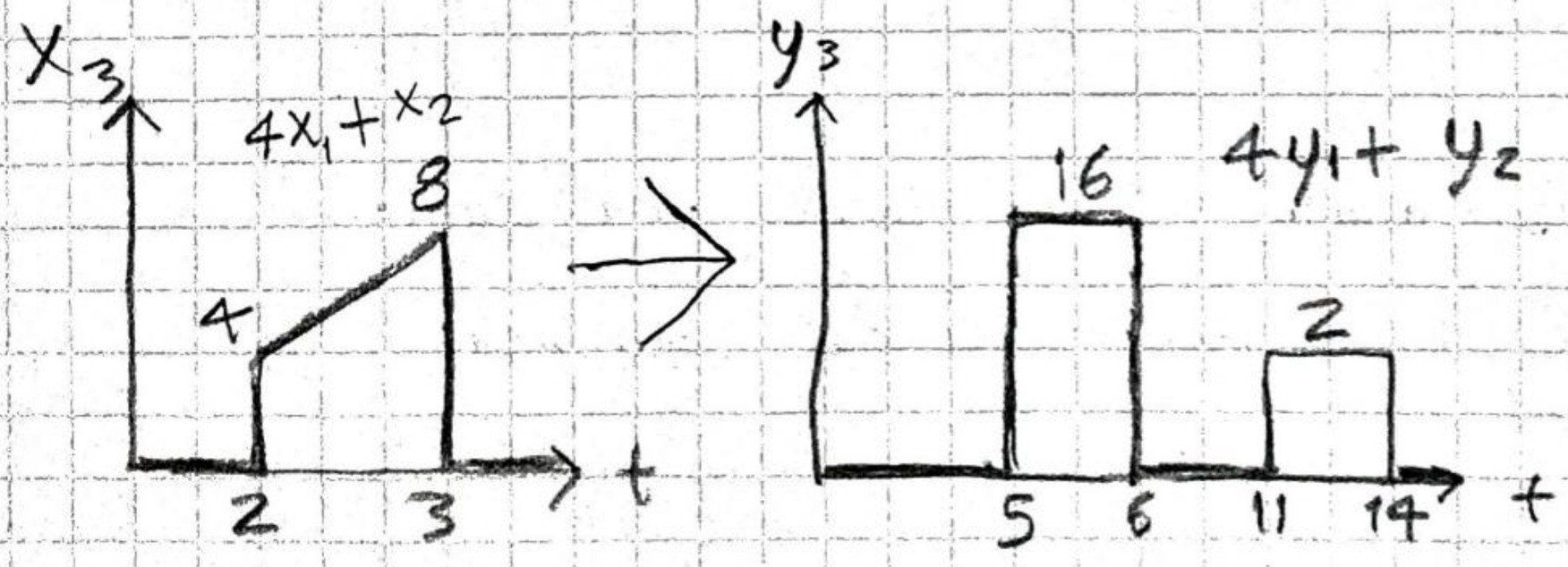
$A x_1 = y_1$ (matr.) $\qquad A x_2 = y_2$

$A(a x_1 + b x_2) = \underbrace{a A x_1 + b A x_2}_{\text{no need}} = a y_1 + b y_2$

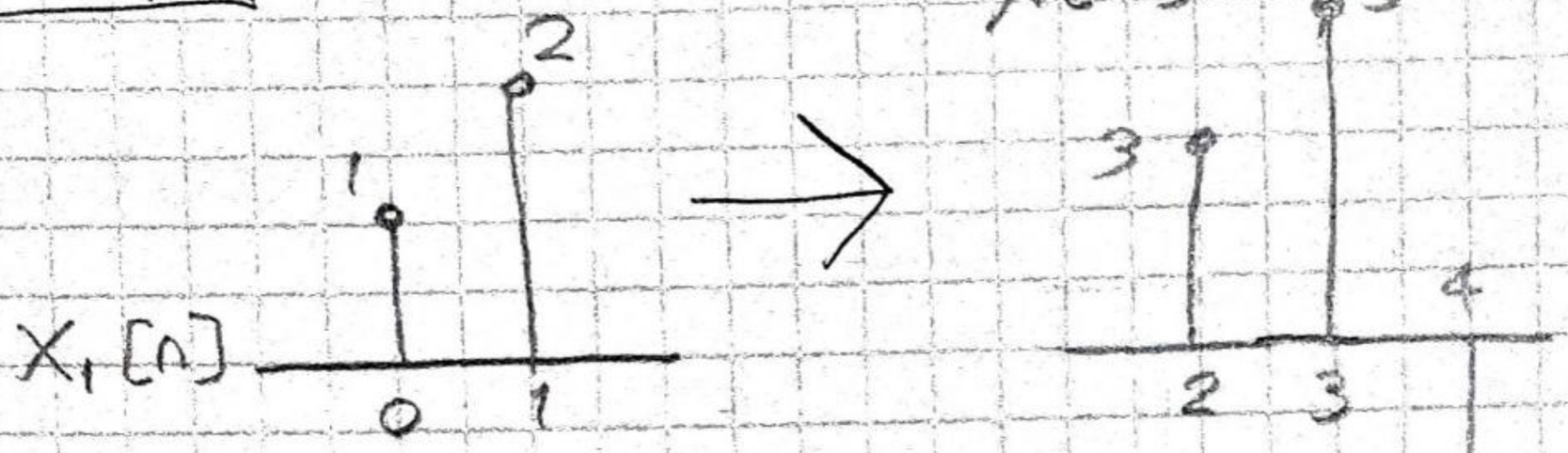
example: ①

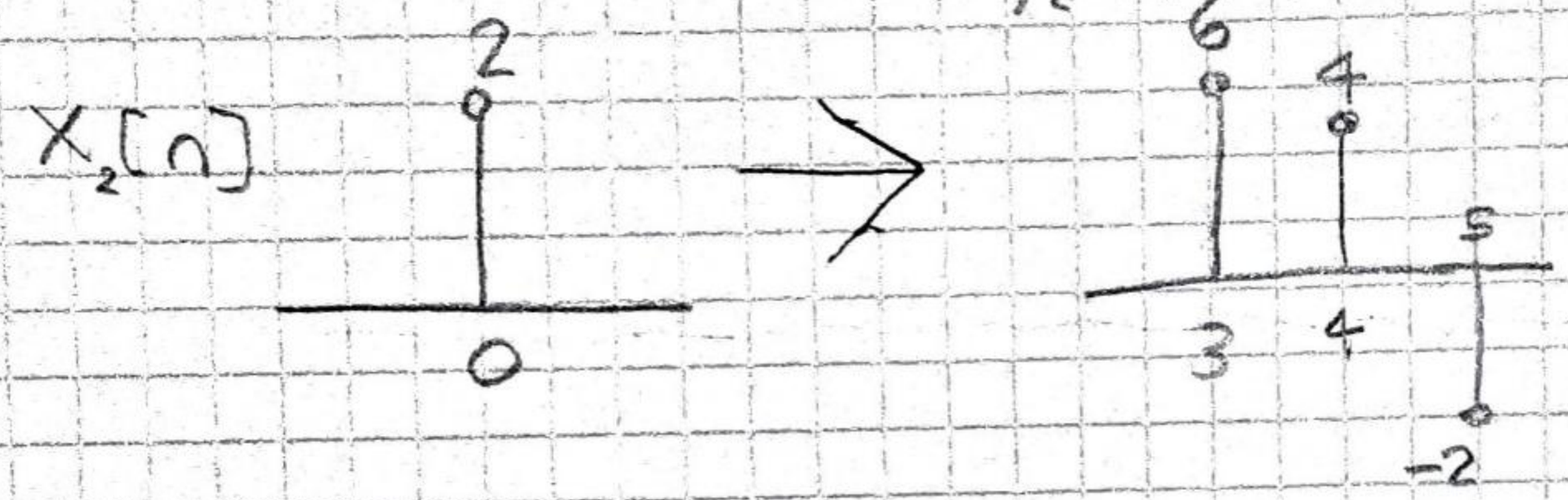




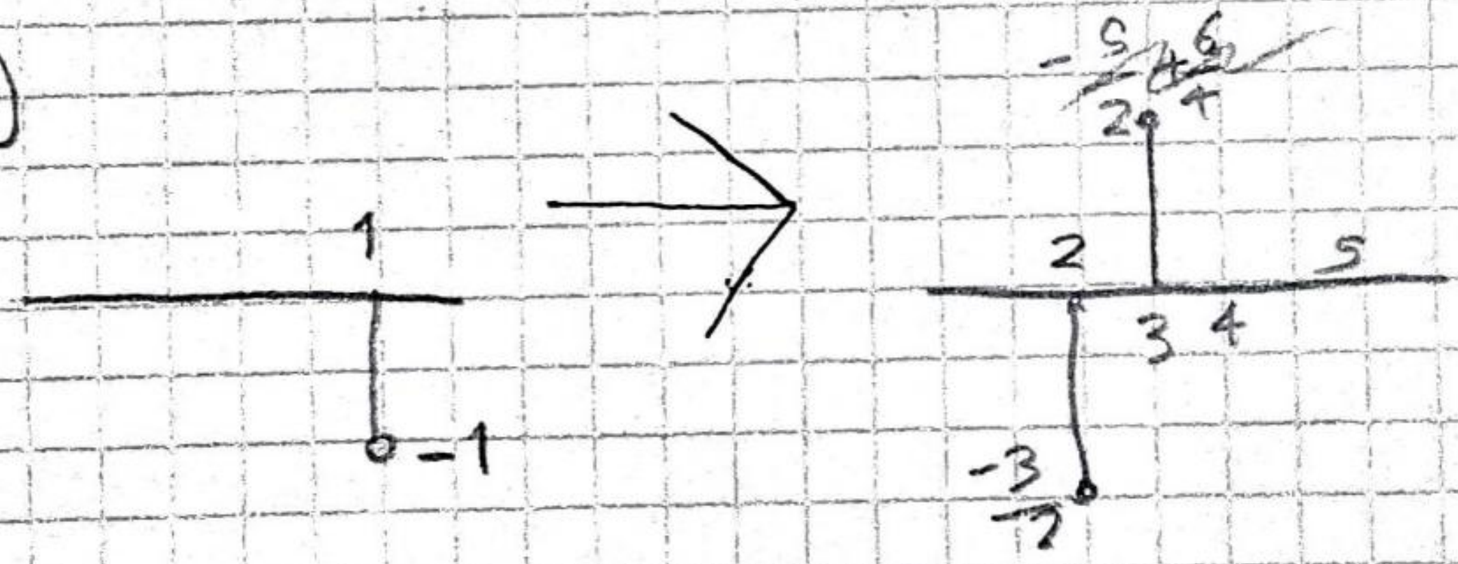


example





$x_{new}[n]$



$$-\frac{1}{2}x_1 + \frac{1}{4}x_2$$

Example

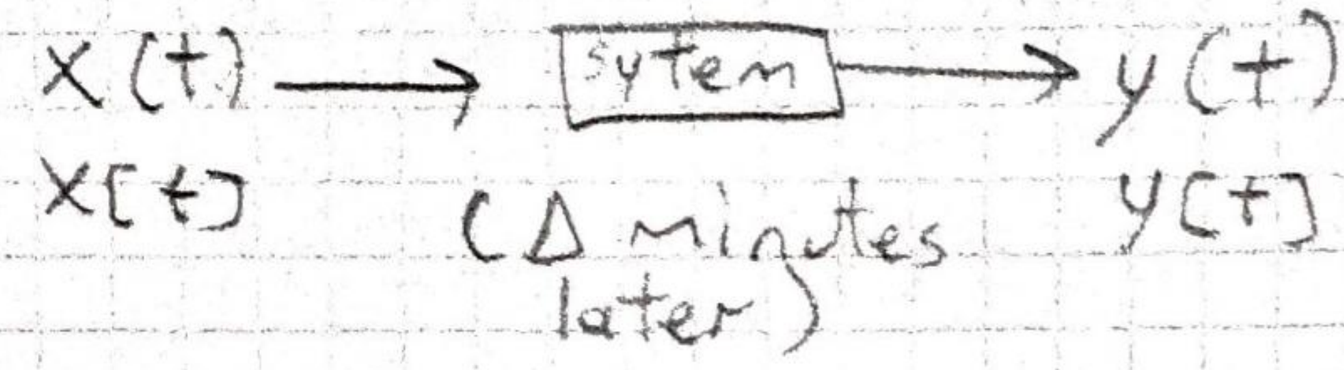
$1 \longrightarrow \boxed{x^2} \longrightarrow 1$

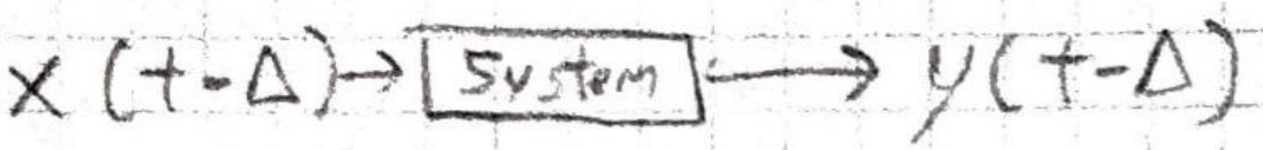
$2 \longrightarrow \boxed{x^2} \longrightarrow 4$

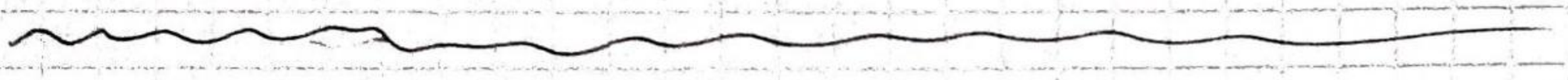
$3 \longrightarrow \boxed{?} \longrightarrow 9$

I expected 5 but got 9 insted. The system is not linear.

Time Invariance







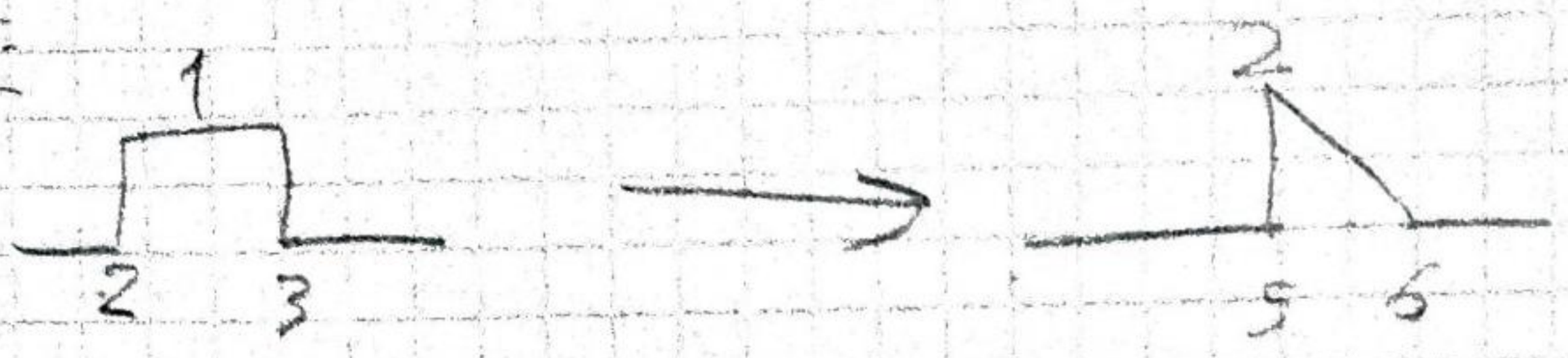
Rule

If a system is LTI and if I know its response $h[n]$ to $\delta[n]$, then I know everything about the system. (ie, I can compute its responses)

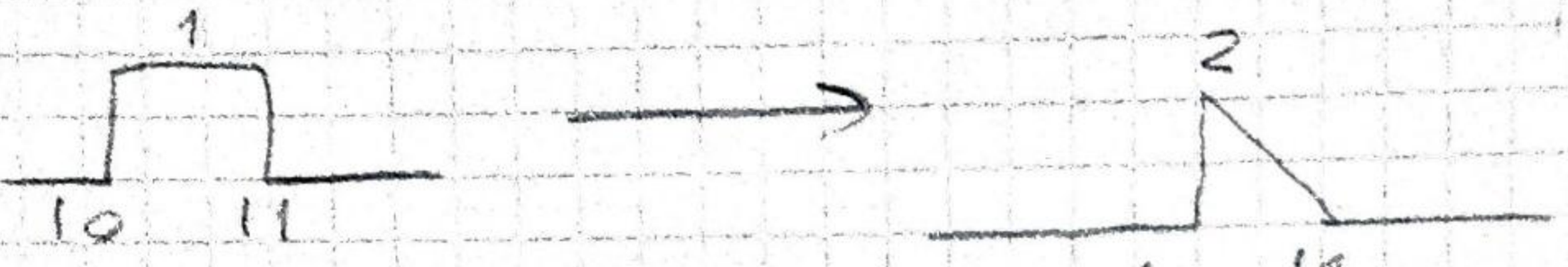
☆ No need to know the internal details.

TIME invariance examples

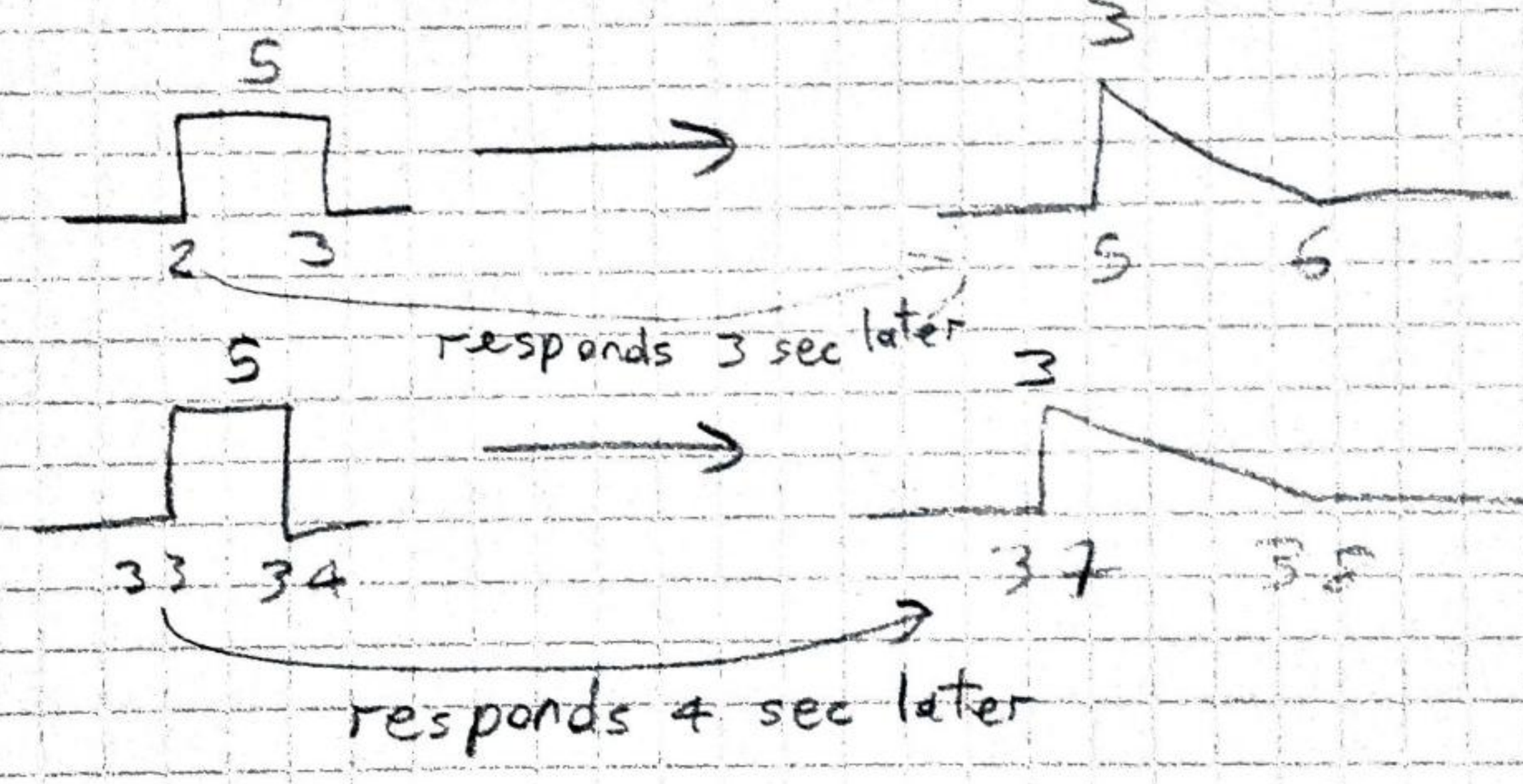
ex 1:



Time invariant



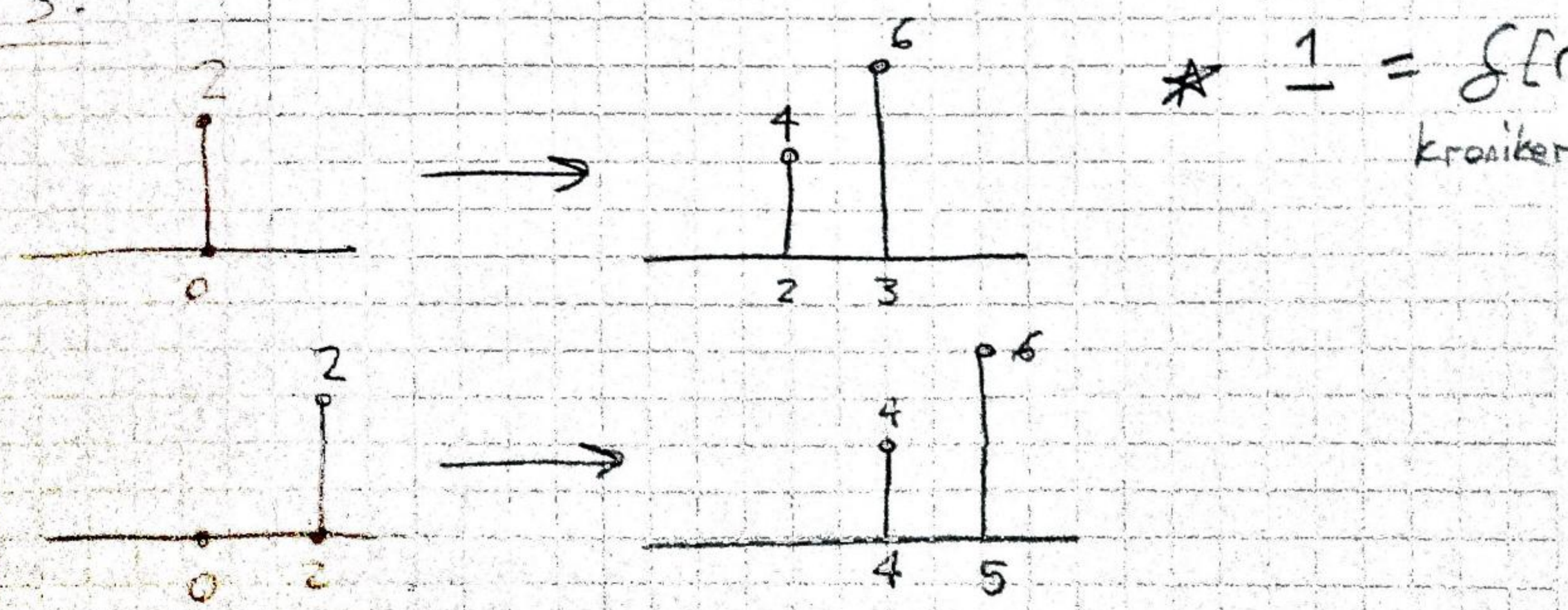
ex 2:



Time variant!

System slowed down. Not time invariant.

ex 3:



* $\frac{1}{}$ = δ[n]

kronicker

* Send kronicker delta to know everything about it without knowing inside. (For LTI system)

$$x[n] = \sum_{k=-\infty}^{\infty} \delta[n-k]\, x[k]$$

shift, scale

ex

2 $\underline{-1}$ 3 4 2 1 0 1

$= 2\delta[n-1] - 1\delta[n] + 3\delta[n-1] + 4\delta[n-2] + 2\delta[n-3]$

$+ \delta[n-4] + \delta[n-6]$

ex

$\delta[n] \longrightarrow \boxed{LTI} \longrightarrow h[n]$

$2\delta[n+1] + \delta[n-1] \longrightarrow \boxed{LTI} \longrightarrow 2h[n+1] + h[n-1]$

$x[n] = \sum x[k]\delta[n-k] \rightarrow \boxed{LTI} \longrightarrow y(n) = \sum x[k]h[n-k]$

Problem

$h[n] = \underline{0}\ 1\ -1\ 2$

$x[n] = 1\ 1\ 2\ 0\ 3$

$y[n] = ?$

for each n, evaluate $\sum_{k=-\infty}^{\infty} x[k]\, h[n-k]$

for $n=0$ $\quad y[0] = \sum_{k=-\infty}^{\infty} x[k]\, h[0-k] = \sum_{k=-\infty}^{\infty} x[k]\, h[-k]$

$h[-k] = 2\ -1\ 1\ \underline{0}$

$x[k] = \quad 1\ 1\ \underline{2}\ 0\ 3$ $\Big\}$ $y[0] = 0$

2 -1 1 0

1 1 2 0 3

$2 - 1 + 2 = 3$

$y[1] = 3$

2 -1 1 0

1 1 2 0 3

$y[2] = -2 + 2 = 0$

2 -1 1 0

1 1 2

$y[-1] = 1$

This is discrete convolution!

example

$x[n] = 2 \quad 1 \quad \underline{0} \quad 1 \quad 2 \rightarrow$ input

$h[n] = \underline{0} \quad 1 \quad -1 \rightarrow$ impulse response

	2	1	0	1	2	
time: -2	-1	1	0			= 0
time: -1		-1	1	0		= 2
time = 0			-1	1	0	= -1
time: 1				-1	1	0 = -1
time = 2					-1	1 0 = 1
time = 3						-1 1 0 = 1
time = 4						-1 1 0 = -2

$y[n] = 0 \quad 2 \quad \underline{-1} \quad -1 \quad 1 \quad 1 \quad -2 \rightarrow$ output

Continues to Convultion...

Signal size S: first non-zero to the last.

..0 0 1 0 1 1 0 1 0 1 0 0 ..

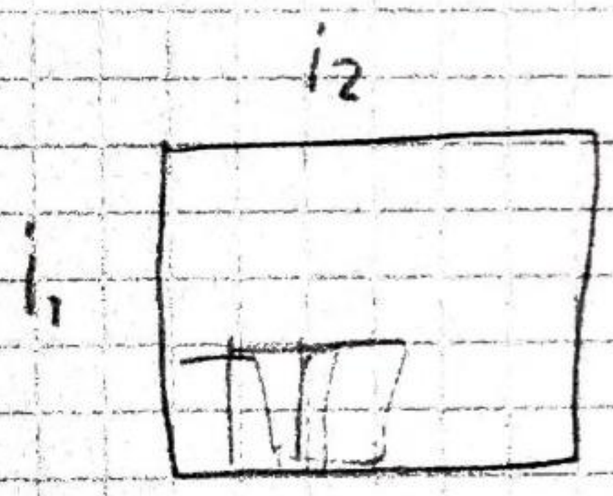
S

filter size f: ..0 1 5 0 0 3 0 ..

f

length of the result: s+f-1

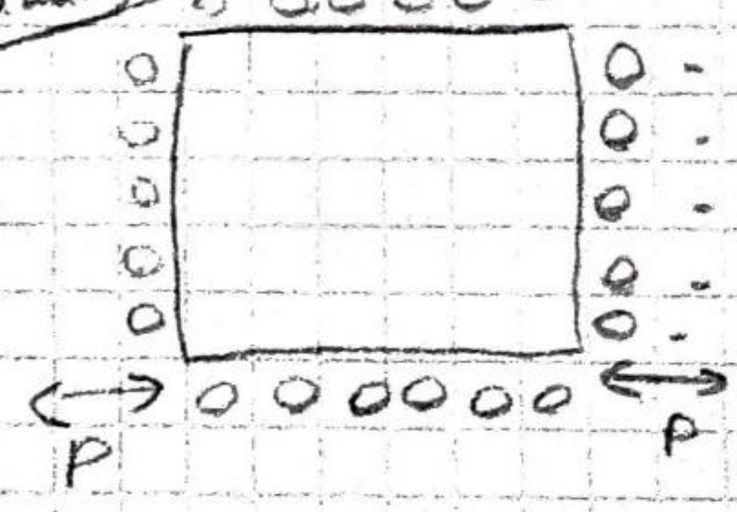
2D Convolution



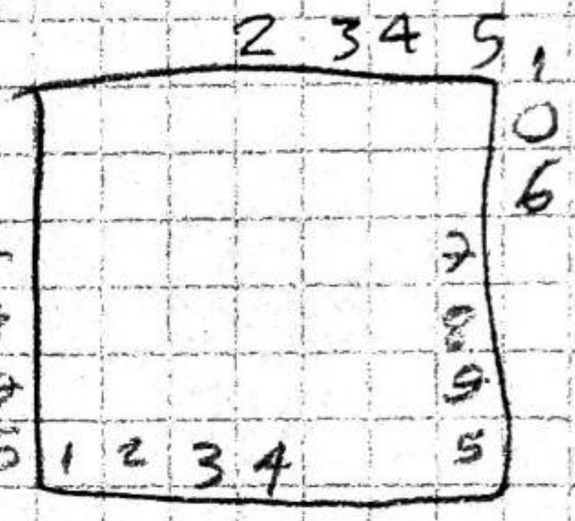
Padding

put many 0 to widen.

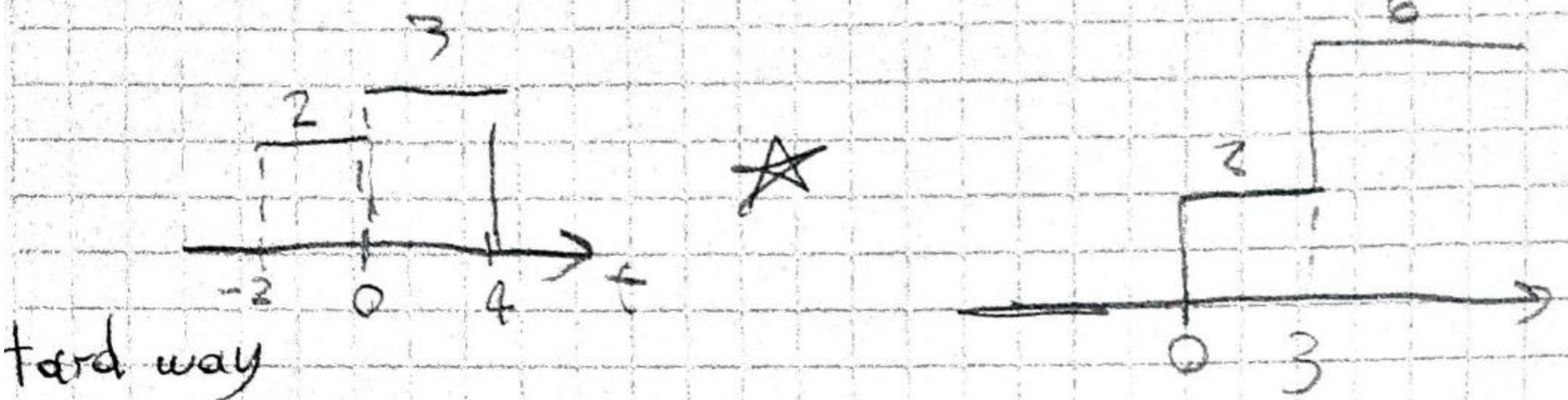
zero Padding



Periodic Padding

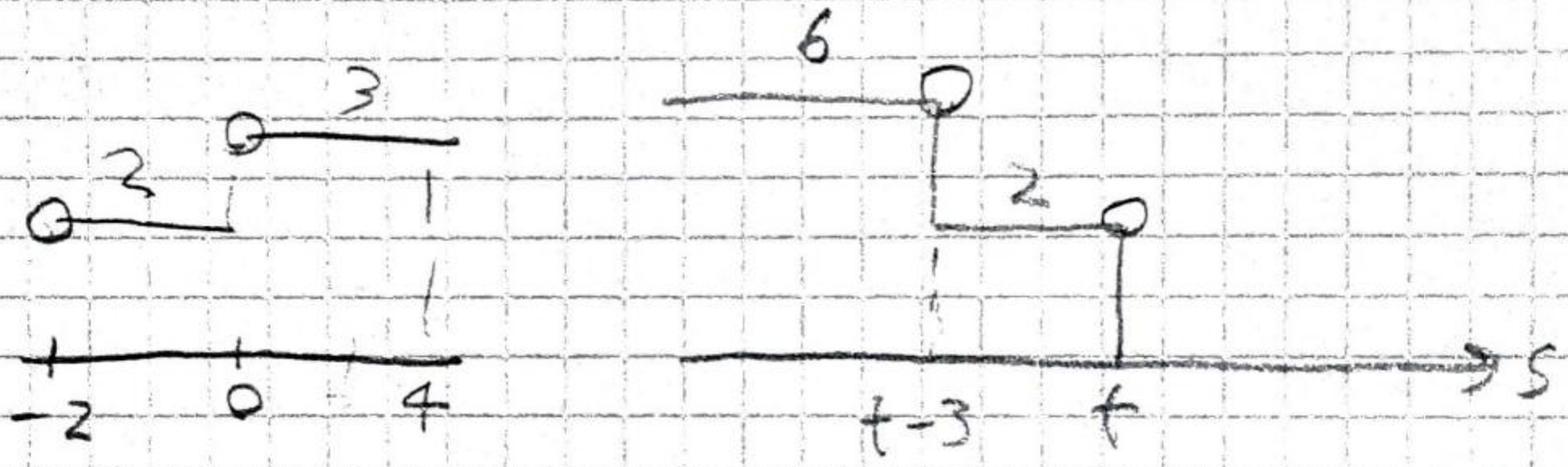


Convolution example



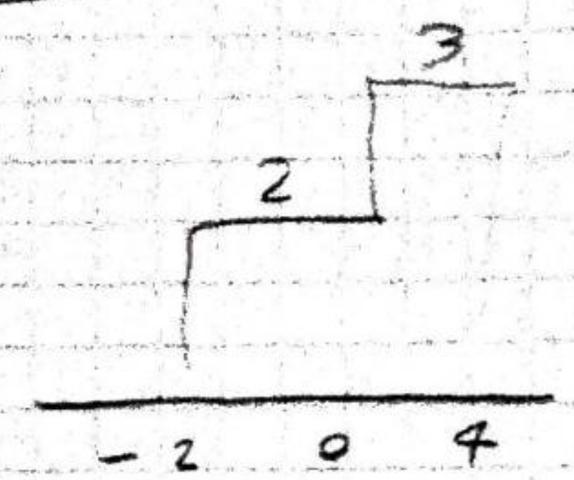
tard way

first: flip and shift. (doesn't matter which to flip in this case)

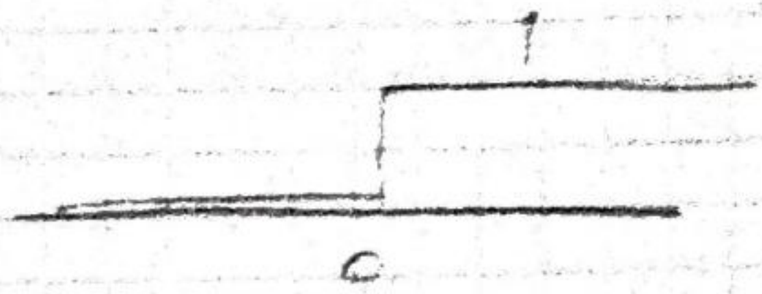


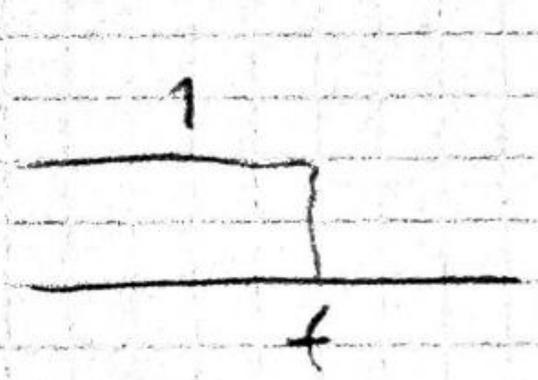
$$
\begin{cases}
0, & t < -2 \\
\int_{-2}^{t} 4\,ds, & -2 < t < 0 \\
\int_{-2}^{0} 2\,ds + \int_{0}^{t} 6\,ds, & 0 < t < 1 \\
\int_{-2}^{t-3} 6 \cdot 2\,ds + \int_{t-3}^{0} 2 \cdot 2\,ds + \int_{0}^{t} 2 \cdot 3\,ds, & 1 < t < 3 \\
\int_{-2}^{0} 6 \cdot 2\,ds + \int_{0}^{t-3} 6 \cdot 3\,ds + \int_{t-3}^{t} 2 \cdot 3\,ds, & 3 < t < 4 \\
\int_{-2}^{t-3} 6 \cdot 2\,ds + \int_{t-3}^{0} 3 \cdot 6\,ds + \int_{0}^{4} 3 \cdot 2\,ds, & 4 < t < 7 \\
\int_{-2}^{0} 6 \cdot 2\,ds + \int_{0}^{t-3} 6 \cdot 3\,ds, & 7 < t
\end{cases}
$$

easy way



finding response





$$\int_{-2}^{t} 2\,ds \qquad -2 < t < 0$$

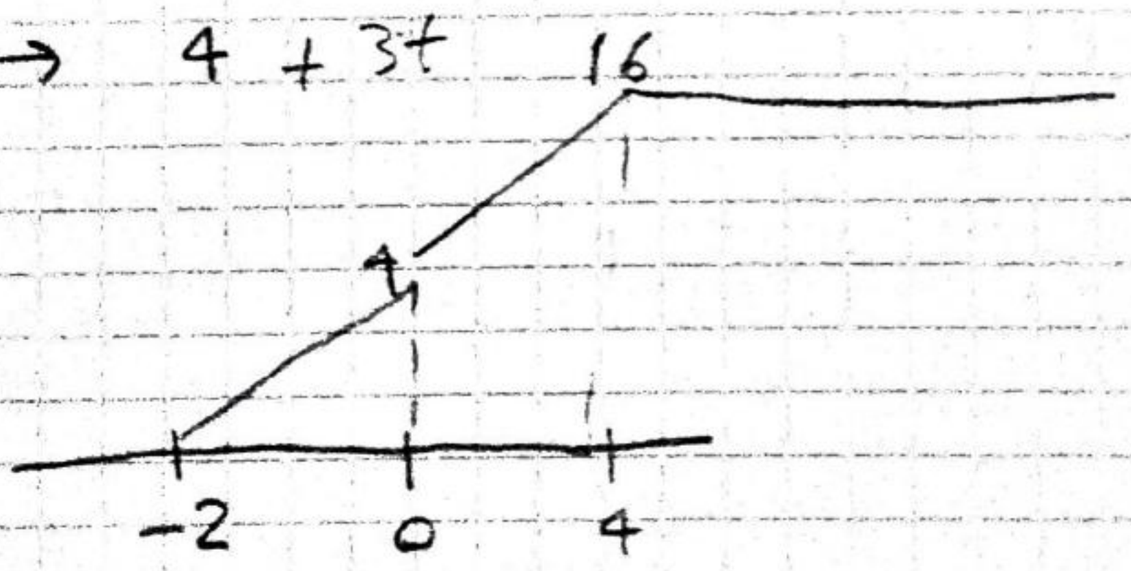
$$\int_{-2}^{0} 2\,ds + \int_{0}^{t} 3\,ds \qquad 0 < t < 4$$

$$\int_{-2}^{0} 2\,ds + \int_{0}^{4} 3\,ds \qquad 4 < t$$

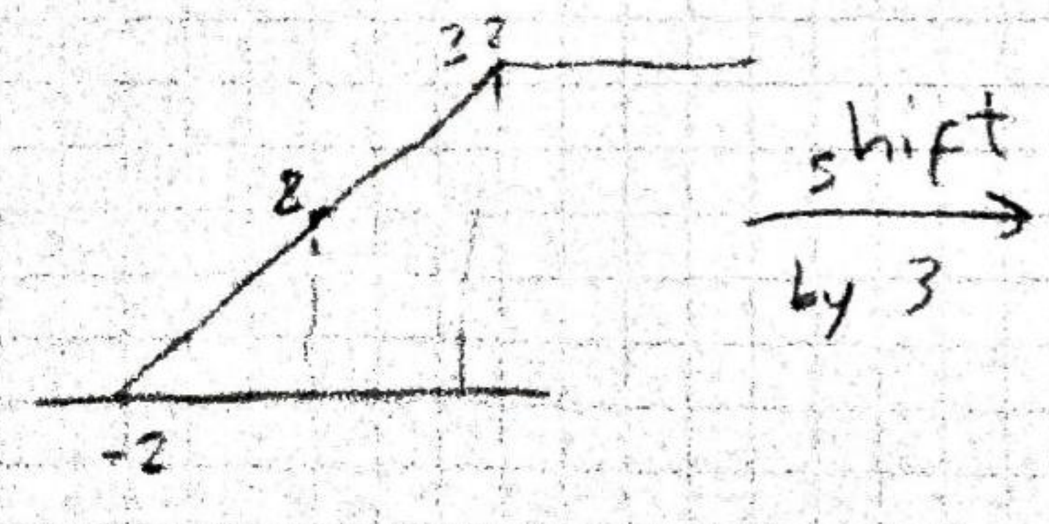
Computing

$-2 < t < 0 \rightarrow 2(t+2)$ (from integral)

$0 < t < 4 \rightarrow 4 + 3t$

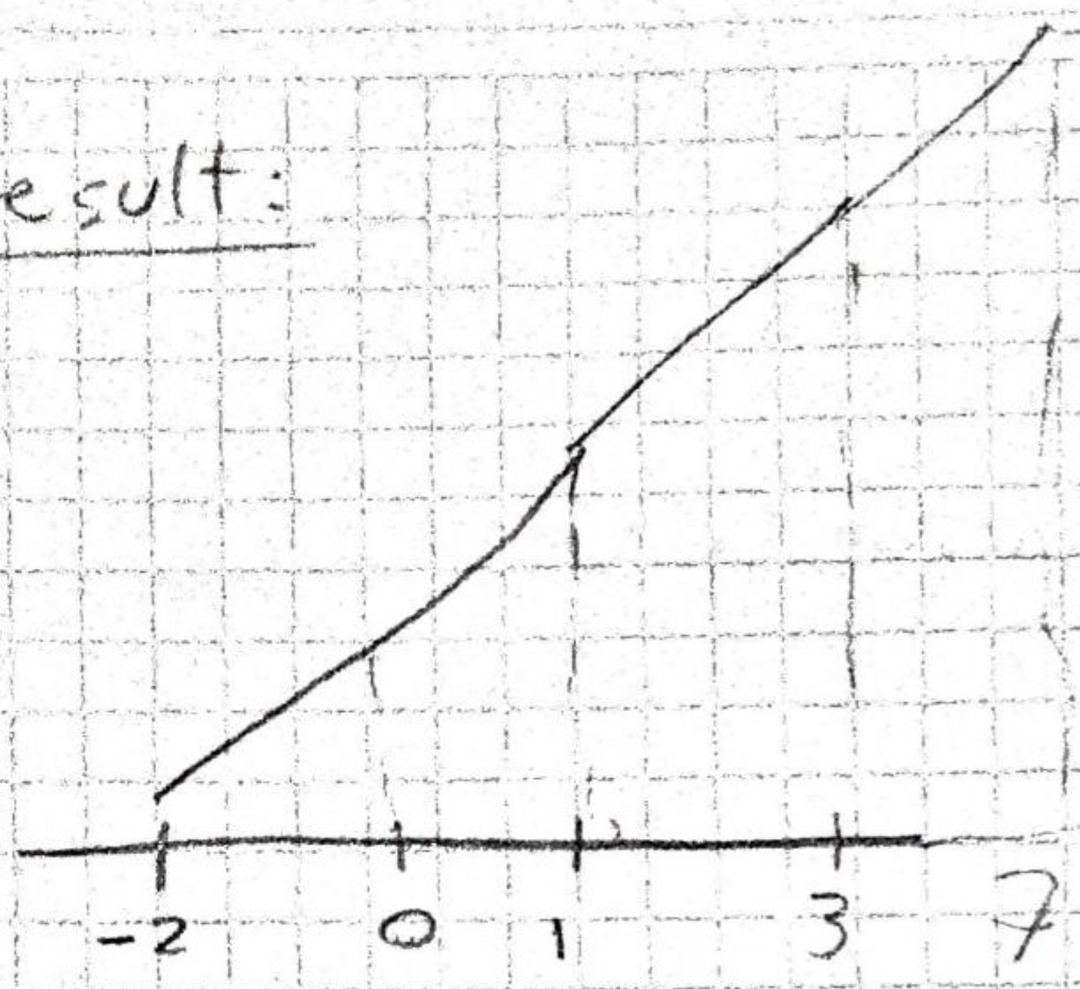


mult. by 2

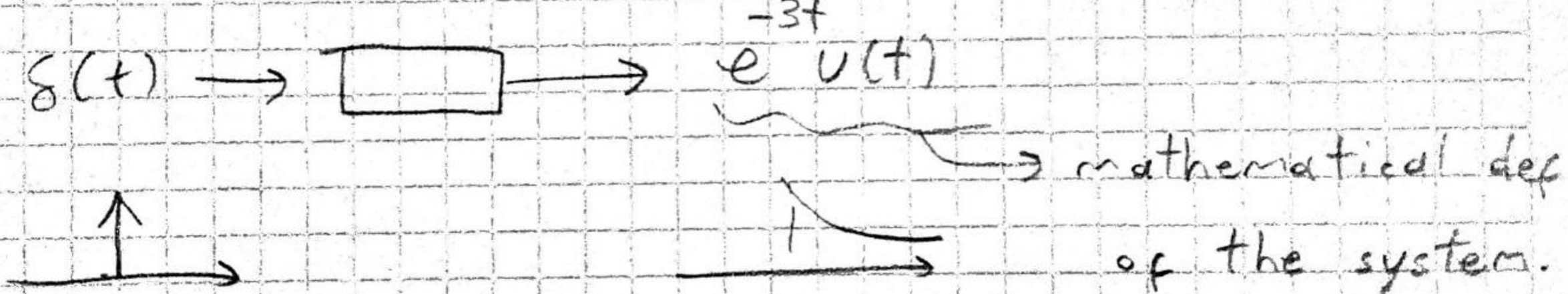


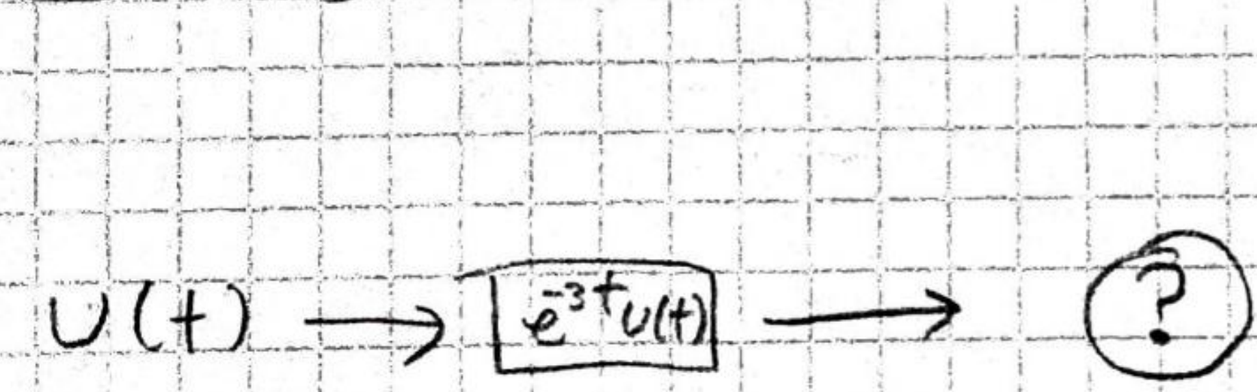
(next page)

result:



Convolution



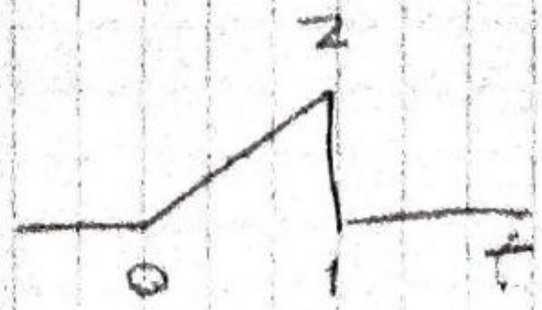


Convolve $u(t)$ with $e^{-3t}u(t)$

$$\int_{-\infty}^{\infty} e^{-3(t-s)} u(t-s) \cdot u(s)\, ds \quad \text{OR} \quad \int_{-\infty}^{\infty} e^{-3s} u(s) \cdot u(t-s)\, ds$$

27.04.2022

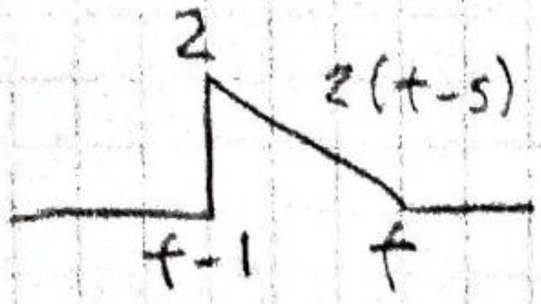
example

Convolve 

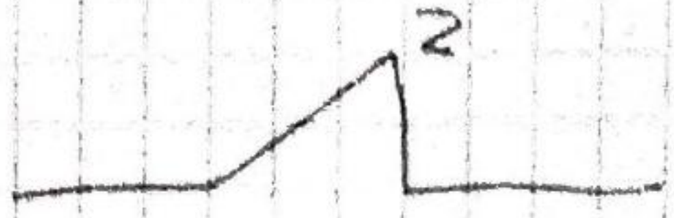
with itself.

$2t\,\chi_{(0,1)}(t)$

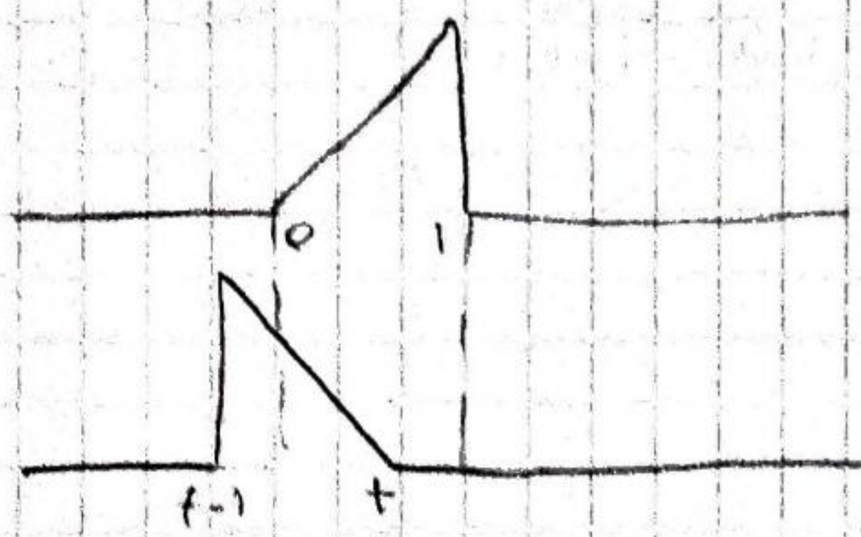
reversed:



$2(t-s)\,\chi_{(0,1)}(t-s)$



$2s\,\chi_{(0,1)}(s)$



$$\begin{cases} 0, & t<0 \\ \int_0^t 2(t-s)2s\,ds & 0<t<1 \\ \int_{t-1}^{1} 2(t-s)2s\,ds & 1<t<2 \\ 0, & 2<t \end{cases}$$

Calculations

$$\int_0^t 4(ts-s^2)\,ds = \left(\frac{4ts^2}{2} - \frac{4s^3}{3}\right)\Big|_0^t$$

$$= 2t^3 - \frac{4t^3}{3} = \frac{2t^3}{3}$$

(bu bir örnekti)

2D Convolution (Discrete)

Choices about boundary cond.

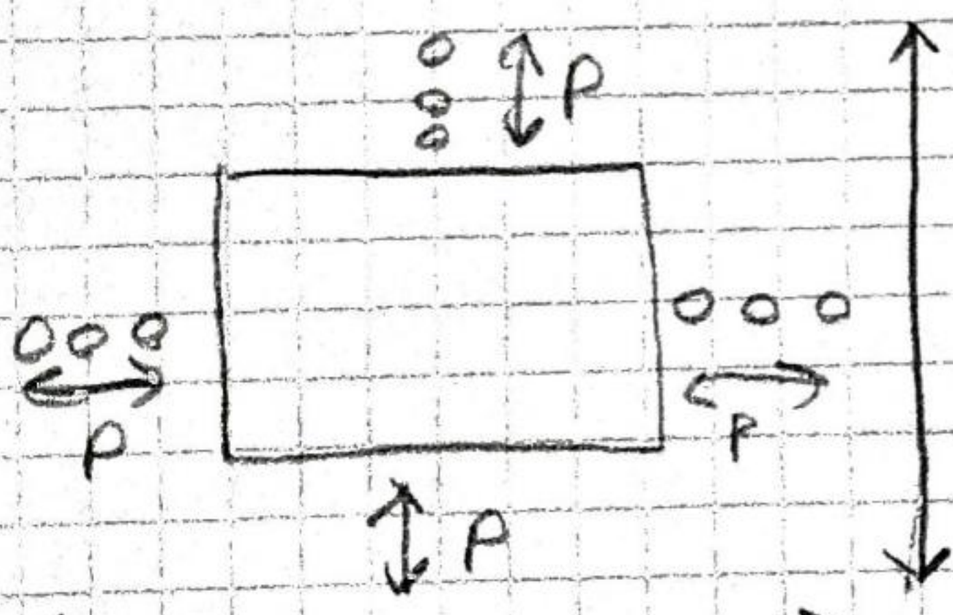
1) never let your filter go out of image

☆ Con: image size will shrink.

$I_1 \times I_2 * F_1 \times F_2 = (I_1 - F_1 + 1, I_2 - F_2 + 1)$

img filter

2) Add zero padding



$(I_1 + 2p - F_1 + 1, I_2 + 2p - F_2 + 1)$

To set the input size = output size: $2p - F + 1 = 0$ olmalı.

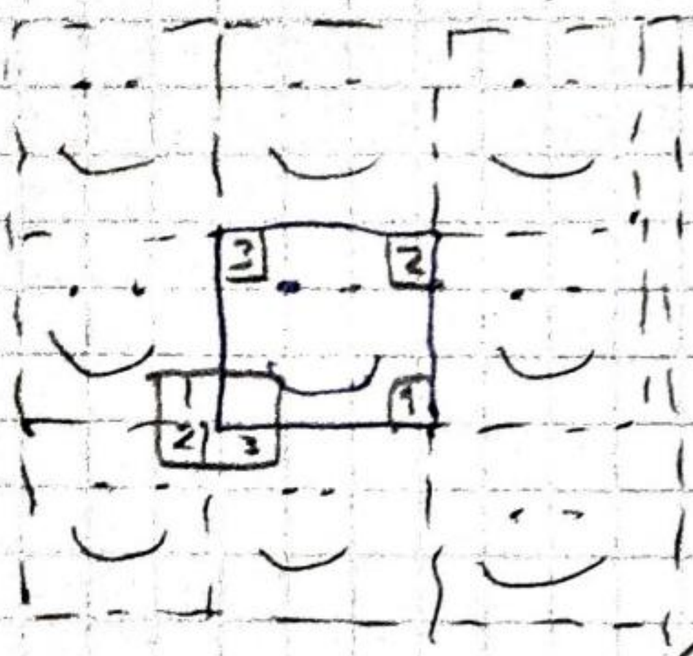
$p = \frac{F-1}{2}$ F çift ise??

Bir tarafa bir sütun daha padding ekle.

☆ Filter arttıkça padding de artar.

☆ F is generally odd

3) Periodic Padding



ex 1: add p=2 zero padding to image

0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	1	2	3	0	0
0	0	4	5	6	0	0
0	0	7	8	9	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0

example continues:

same image, add p=2 periodic padding.

answer:

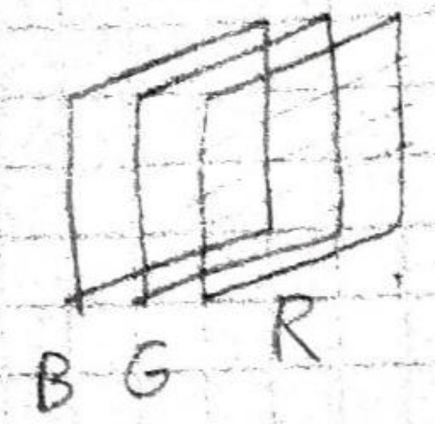
8	9	7	8	9	7	8
2	3	1	2	3	1	2
5	6	4	5	6	4	5
8	9	7	8	9	7	8
2	3	1	2	3	1	2
5	6	4	5	6	4	5

4) Stride

filter'ı kaydırma miktarı: stride

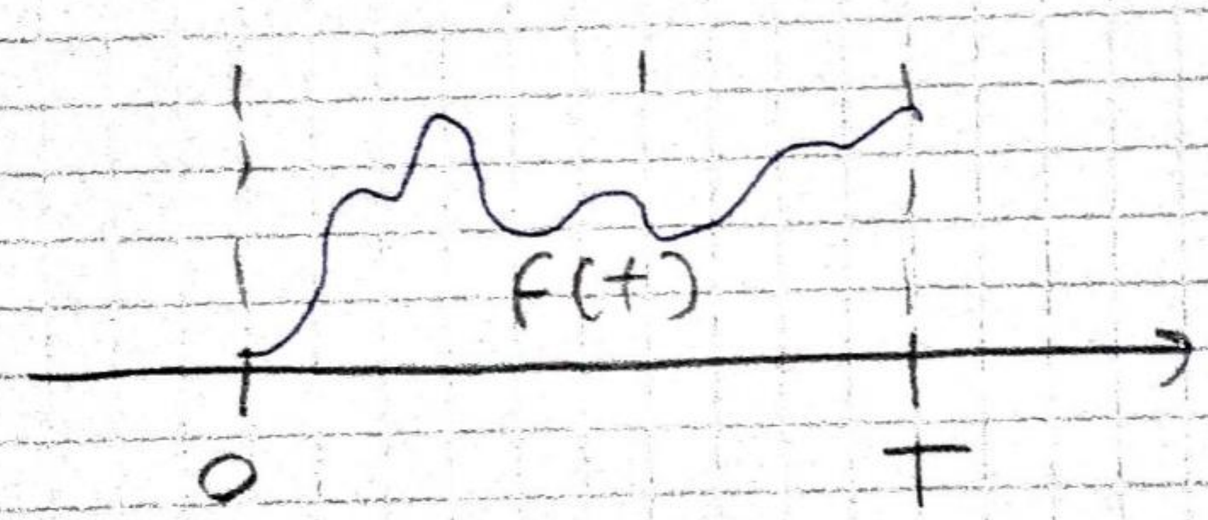
$$result = \left[\frac{I - F + 2P}{S} + 1 \right]$$

5) Channels



A num of channels in the filter should be equal to the image's.

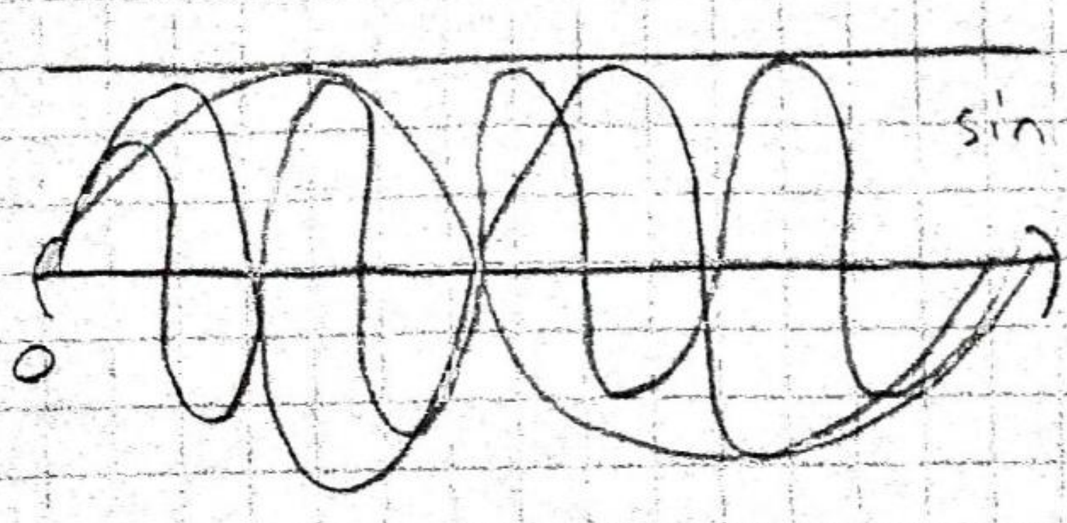
Fourier Transforms



$$f(t) = a_0 \cdot 1 + a_1 \sin\left(\frac{2\pi}{T}t\right) + a_2 \sin\left(2\frac{2\pi}{T}t\right) + a_3 \sin\left(3\cdot\frac{2\pi}{T}t\right)$$

$$\ldots + b_0 \cos\left(\frac{2\pi}{T}t\right) + b_1 \cos\left(\frac{}{T}t\right) + b_2 \cos\left(3\frac{2\pi}{T}t\right)$$

...



$\sin\left(\frac{3\pi}{T}t\right)$, $\sin\left(2\frac{2\pi}{T}t\right)$

$\sin\left(3\frac{2\pi}{T}t\right)$

then cosines...

→ is it possible to find a sum of thes that is equal to $f(t)$ arbitrary function?

28.04.2022

(Fourier cont.)

norm

$$v = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} \qquad \|v\|_2 = \sqrt{a_1^2 + a_2^2 + a_3^2} = \sqrt{\sum_{i=1}^{3} a_i^2}$$

$$\|v\|_1 = \sum_{i=1}^{3} |a_i|$$

angle (btw two vectors)

$$v_1 \cdot v_2 = \|v_1\| \|v_2\| \cos\theta \qquad \theta = \operatorname{acos}\left(\frac{v_1 \cdot v_2}{\|v_1\| \|v_2\|}\right)$$

example $\quad v_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad v_2 = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix}$

$\|v_1\|_2 = ?$

$\|v_2\| = ?$

$\sqrt{14}$

$\sqrt{14}$

$v_1 \cdot v_2 = 1 \cdot 2 + 2 \cdot 3 + 3(-1) = 5$

$$\theta = \operatorname{acos}\left(\frac{5}{14}\right)$$

Uncountably Infinite Version

norm

$$\|f(x)\|_2 = \sqrt{\int_{-\infty}^{\infty} f(x)^2 \, dx}$$

for finite ↓

angle

$f(x), g(x)$

$$\cos\theta = \frac{v_1 \cdot v_2}{\|v_1\| \|v_2\|}$$

for infinite ↓

$$\frac{\int_{-\infty}^{\infty} f(x) g(x) \, dx}{\sqrt{\int f(x)^2 dx} \sqrt{\int g(x)^2 dx}}$$

Orthonormality

- all vectors have the length of 1.

$[\|e_1\| = 1, \|e_2\| = 1, \ldots, \|e_n\| = 1$

- all are orthogonal to each other. (their dot prod. must be kroniker delta)

$i \neq j$. $e_i \cdot e_y = \underbrace{\|e_i\|}_{1}\underbrace{\|e_j\|}_{1}\underbrace{\cos\theta_{ij}}_{\cos 90} = 0$

(dot product)

$$\boxed{e_i \cdot e_j = \delta_{ij}} \leftarrow \text{orthonormality}$$

dot kroniker

★ Then I can express any v in terms of e_i.

$$v = (v \cdot e_1)e_1 + \underbrace{(v \cdot e_2)}_{\text{num}}e_2 + \ldots + \underbrace{(v \cdot e_n)}_{\text{num}}e_n$$

dot prod. → vec; num → vec; num → vec

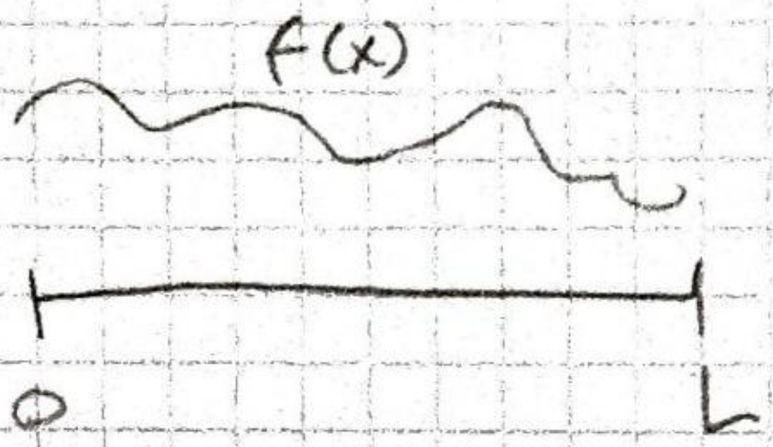
ex $v = 3e_1 + 5e_2 + 9e_3$

$3 = ve_1$ $5 = ve_2$ $9 = ve_3$

Terminalogy: we expand V in $e_1 \ldots e_n$

Fourier: we expand $f(x)$ in sin, cos.

Fourier Series



$$f(x) = a + \sum^{\infty} b_k \sin\left(\frac{2\pi k}{L}x\right) + \sum^{\infty} c_k \cos\left(\frac{2\pi k}{L}x\right)$$

$k=0$

e_k

$v \cdot e_k$

like in $v = \sum (v e_k) e_k$

$\star \quad \frac{2\pi}{L} = \omega$

$$\boxed{f(x) = a + \sum_{k=0}^{\infty} b_k \sin(k\omega x) + \sum_{k=0}^{\infty} c_k \cos(k\omega x)}$$

$$= a + \sum_{k=0}^{\infty} b_k \frac{e^{ik\omega x} - e^{-ik\omega x}}{2i} + \sum_{k=0}^{\infty} c_k \frac{e^{ik\omega x} + e^{-ik\omega x}}{2}$$

$$\boxed{f(x) = \sum_{k=-\infty}^{\infty} d_k e^{ik\omega x} \qquad d_{-k} = d_k^{\star}}$$

→ orthanormal (prove)

→ number

→ $e^{i\omega k x} \cdot e^{i\omega m x} \stackrel{?}{=} \delta_{nm}$

$$\int_0^L e^{i\omega k x} e^{i\omega m x} dx \stackrel{?}{=} \delta_{nm} \qquad \omega = \frac{2\pi}{L}$$

$$\int_0^L e^{i\frac{2\pi}{L}kx} e^{i\frac{2\pi}{L}mx} dx \stackrel{?}{=} \delta_{nm}$$

If i can prove this, $e^{i\frac{2\pi}{L}x}$ is an orthonormal basis.

Basis (orthonormal)

$$V = \alpha_1 e_1 + \alpha_2 e_2 + \dots + \alpha_n e_n \longrightarrow \sum_{i=1}^{n} \alpha_i e_i$$

$\alpha_1, \alpha_2, \alpha_n$?

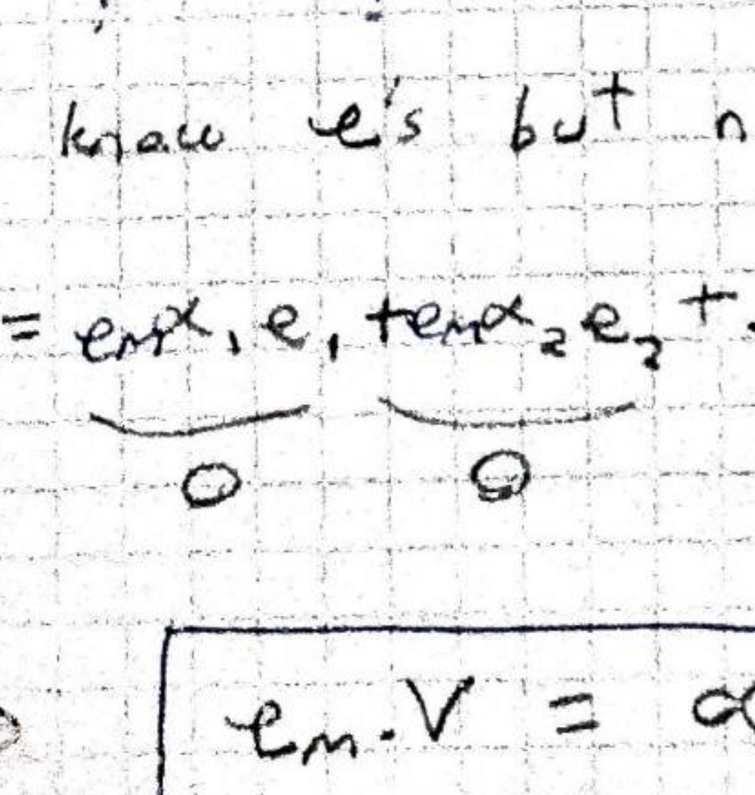
we know e's but not α's.

$$e_m V = e_m \alpha_1 e_1 + e_m \alpha_2 e_2 + \dots + e_m \alpha_m e_m + \alpha_n e_n \cdot e_m$$

$e_m \alpha_1 e_1$: 0 — $e_m \alpha_2 e_2$: 0 — $e_m \alpha_m e_m$: 1 — $\alpha_n e_n \cdot e_m$: 0

$$\boxed{e_m \cdot V = \alpha_m}$$

example



given:

$|V| = 43$ in terms of e?

$V \cdot e_1 = |V||e_1| \cos(60) = 43 \cdot \frac{1}{2}$

$V \cdot e_2 = 0$

$V \cdot e_3 = 43 \cos 30$

$$V = \frac{43}{2} e_1 + 0 \cdot e_2 + \frac{43\sqrt{3}}{2} e_3$$

Consider only functions described in a finite interval.

Dot product of $f(t)$ and $g(t)$:

$$\langle f(t), g(t) \rangle = \int f(t) \overrightarrow{g^*(t)}\, dt$$ → complex conjugate

★ dot product with itself (and sqrt) = length.

$$|f(t)| = \sqrt{\langle f(t), f(t) \rangle} = \sqrt{\int f(t) f^*(t) dt}$$

★ complex conjugate: eşlenik. imaginary kısmı yok eder.

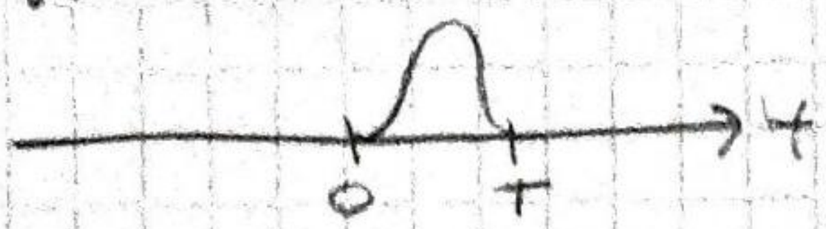
$$\cos\theta = \frac{\langle f(t), g(t) \rangle}{|f(t)|\,|g(t)|}$$

I want to express my func. in a different ort. basis.
I need to find an ort. basis first.

Continuous FS (Fourier series)

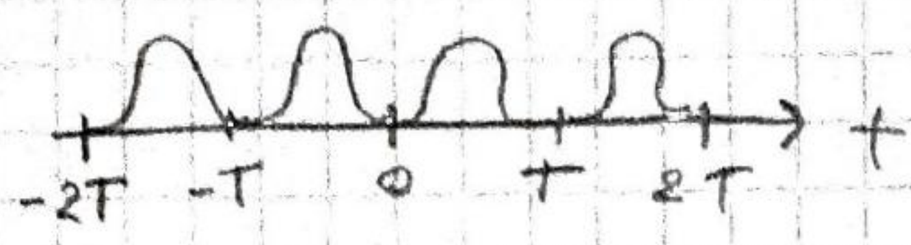
↳ Limited to periodic signals or signals defined on finite interval

finite int.:



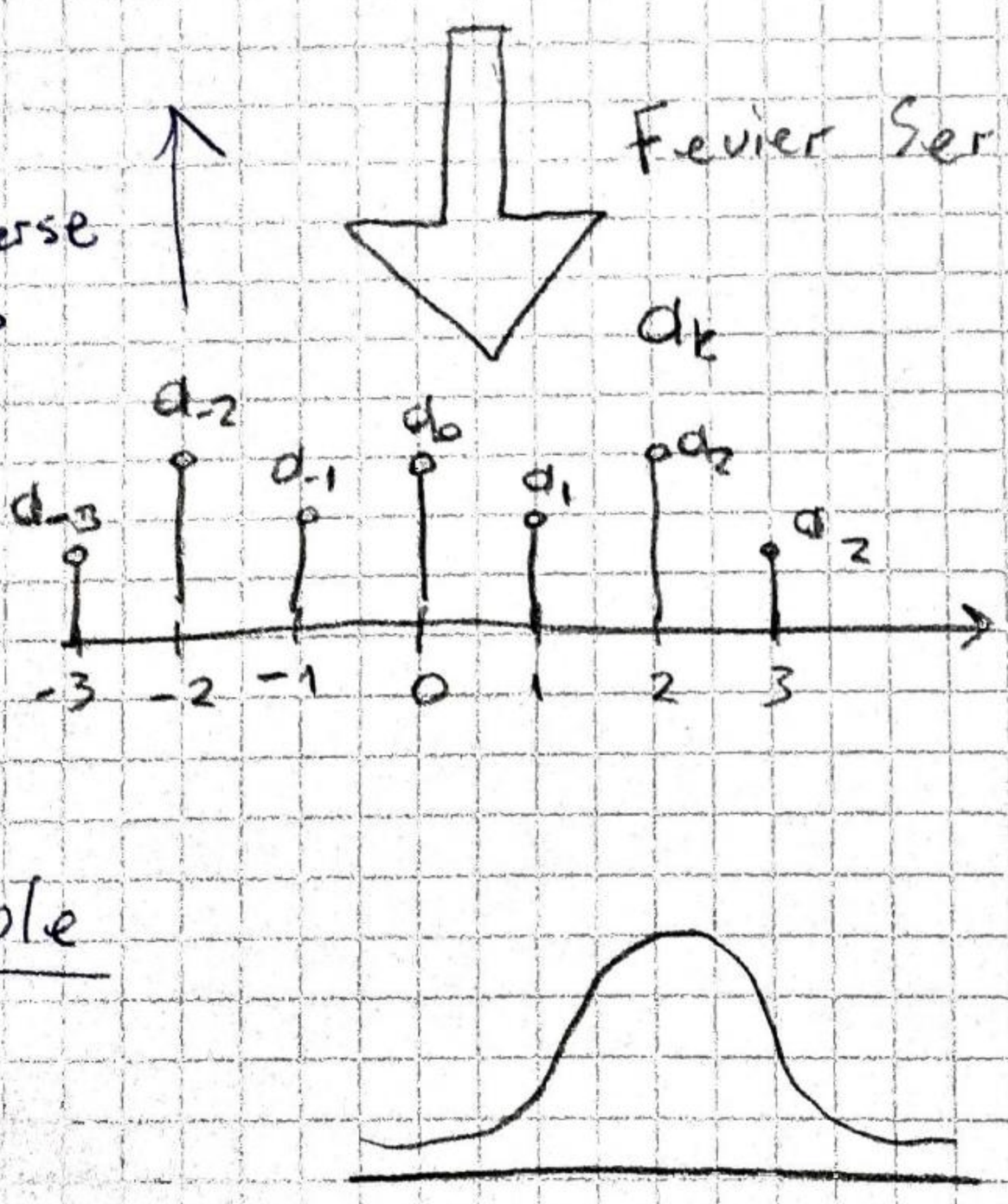
→ Case 1: only interested in 0 to T.

periodic:



→ Case 2: periodic

★ Both cases are equal for us



We take a periodic function or a function that is restricted to a certain interval. And we convert it to a discrete function

↓

FS

example

We can't find FS for this thing, instead we have Fourier transforms.

Fourier Transforms

Signals not necessarily periodic, defined on the whole real line.

We extend FS formula to apply this problem.

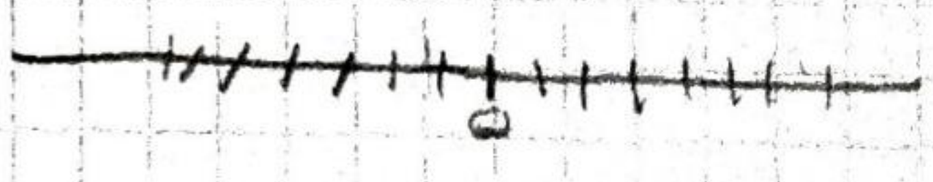
$$\frac{2\pi}{T} = \omega_0$$

→ precision

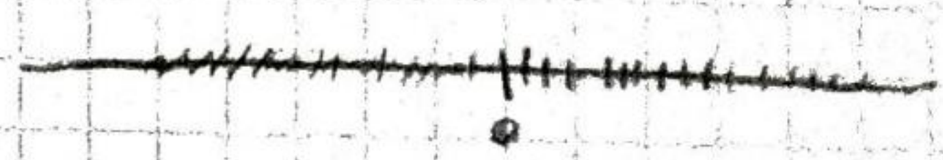
$\omega_0 = 1$

becomes more precise

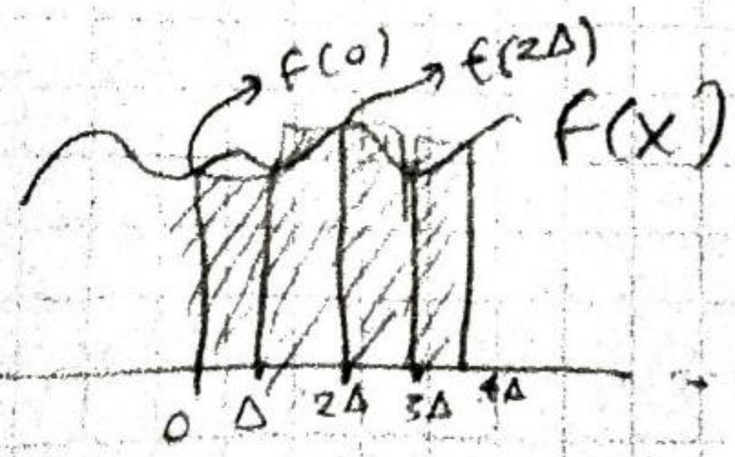
$\omega_0 = 0.25$



$\omega_0 = 0.025$



ASIDE: (ek not)



$$\text{area} \sim \sum_{k=-\infty}^{\infty} F(k\Delta)\Delta$$

$$\lim_{\Delta \to 0} \sum_{k=-\infty}^{\infty} F(k\Delta)\Delta \longrightarrow \int_{-\infty}^{\infty} F(x)\,dx$$

Cont...

$$X(w) = \int_{-\infty}^{\infty} x(t)\, e^{-jwt}\, dt \qquad FT$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(w)\, e^{jwt}\, dt$$

RECAP

- Fourier Series (only for periodic)

$$x(t) = x(t+T)$$

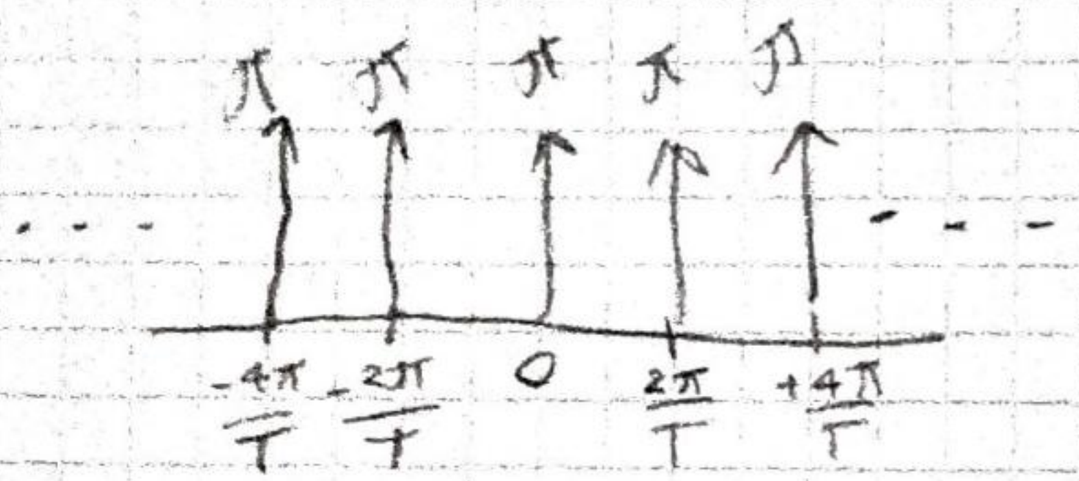
$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{\left(\frac{2\pi}{T}\right)kt}$$

$$a_k = \frac{1}{T} \int_{\langle T \rangle} f(t)\, e^{-\left(\frac{2\pi}{T}\right)kt}\, dt$$

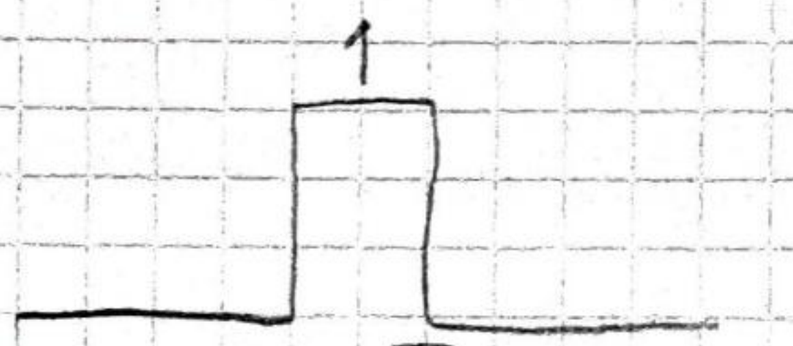
- FT (for periodic and non-periodic)

$$x(t) = \int_{-\infty}^{\infty} X(w)\, e^{jwt}\, dt \qquad IFT$$

$$X(w) = \frac{1}{2\pi} \int_{-\infty}^{\infty} x(t)\, e^{-jwt}\, dt \qquad FT$$



Example



FT of this?

first find a_k (FS coefficients)

$$a_k = \frac{1}{T}\int_{-T_1}^{T_1} 1 \cdot e^{-i\frac{2\pi}{T}kt}\,dt$$

$$a_k = \frac{1}{T}\left.\frac{e^{-i\frac{2\pi}{T}kt}}{-i\frac{2\pi}{T}k}\right|_{-T_1}^{T_1} = \frac{1}{T}\cdot\frac{e^{i\frac{2\pi}{T}kT_1} - e^{-i\frac{2\pi}{T}kT_1}}{-i\frac{2\pi}{T}k}$$

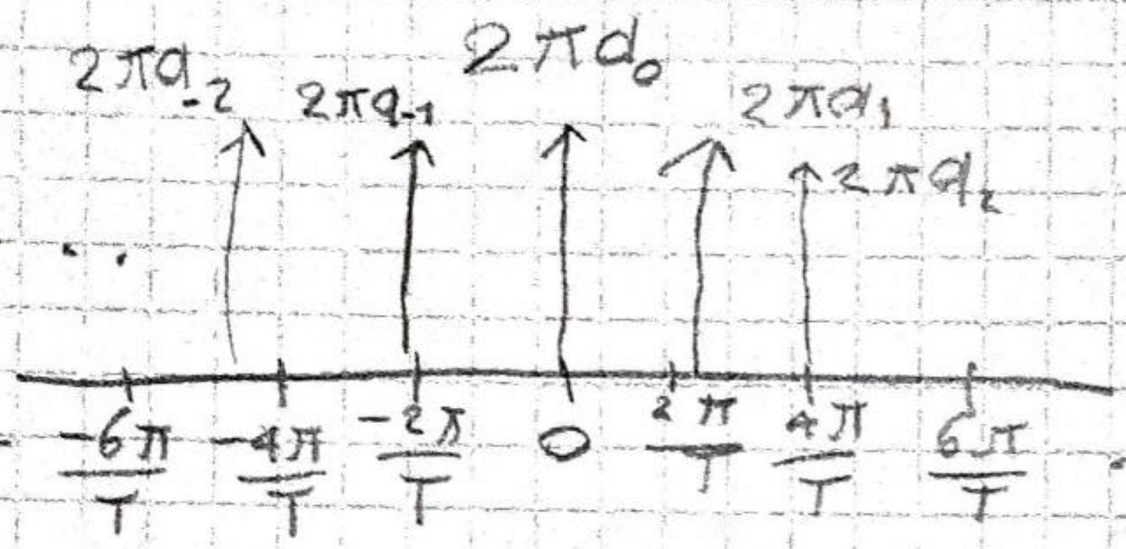
$$= \frac{1}{k\pi}\cdot\frac{e^{i\frac{2\pi}{T}kT_1} - e^{-i\frac{2\pi}{T}kT_1}}{2i} = \frac{1}{k\pi}\sin\left(\frac{2\pi}{T}kT_1\right)$$

Now FT:

$$X(w) = \sum \underbrace{\frac{1}{k\pi}\sin\left(\frac{2\pi}{T}kT_1\right)}_{a_k}\cdot\delta\left(w - \frac{2\pi k}{T}\right)$$

(sonraki sayfada çizimi var

FT'nin çizimi:



if T↑ dirac deltas squeez.

example 2

find FT(δ(t))
(FT of dirac delta)

$$\int_{-\infty}^{\infty} \boxed{\delta(t)}\, e^{-j\omega t}\, dt$$ (dirac delta kills integral)

t=0 for dirac delta

$= 1$

example 3

FT(δ(t-2)) ?

now t=2

$= e^{-j\omega 2}$

example 4

FT[$e^{-2t}U(t)$] ?

$$\int_{-\infty}^{\infty} \boxed{e^{-2t}U(t)}\, e^{-j\omega t}\, dt \rightarrow \int_{0}^{\infty} e^{-(2+j\omega)t}\, dt = \frac{e^{-(2+j\omega)t}}{-(2+j\omega)}\Bigg|_0^{\infty}$$

$$= \frac{e^{-(2+j\omega)\infty} - 1}{-(2+j\omega)} = \frac{1}{2+j\omega}$$

example 5 $FT\left[e^{-2|t|}\right]$

$$\int_{-\infty}^{0} e^{2t}\, e^{-j\omega t}\, dt + \int_{0}^{\infty} e^{-2t}\, e^{-j\omega t}\, dt$$

$$= \frac{e^{(2-j\omega)t}}{2-j\omega}\Bigg|_{-\infty}^{0} + \frac{e^{-(2+j\omega)t}}{-(2+j\omega)}\Bigg|_{0}^{\infty}$$

example 6 $FT(1)$

$$\int_{-\infty}^{\infty} 1 \cdot e^{-j\omega t}\, dt \rightarrow \frac{e^{-j\omega t}}{-j\omega}\Bigg|_{-\infty}^{\infty} = \frac{e^{-j\omega\infty} - e^{j\omega\infty}}{-j\omega}$$

doesn't mean anything!

★ instead:

$$\lim_{T\to\infty} \int_{-T}^{T} e^{-j\omega t}\, dt$$

$$= \lim_{T\to\infty} \frac{e^{-j\omega t} - e^{j\omega T}}{-j\omega} \qquad j = i$$

(similar to sin)

$$= 2 \lim_{T\to\infty} \frac{1}{\omega} \frac{e^{j\omega T} - e^{-j\omega T}}{2j}$$

$$= 2 \lim_{T\to\infty} \frac{\sin(\omega T)}{\omega} = 2 \underbrace{\lim_{\varepsilon\to 0} \frac{\sin(\omega/\varepsilon)}{\omega}}_{\pi\delta(\omega)} = 2\pi\delta(\omega)$$

example 7

$$FT[e^{-\alpha t}U(t)] \quad \alpha > 0$$

$$\int_{-\infty}^{\infty} e^{-\alpha t}U(t)\,e^{-j\omega t}\,dt \rightarrow \int_{-\infty}^{\infty} e^{-(j\omega+\alpha)t}\,dt$$

$$= \frac{e^{-(\alpha+j\omega)t}}{-(j\omega+\alpha)}\Bigg|_0^{\infty} = \frac{e^{-\alpha\infty}\,e^{-j\omega\infty}}{-(\alpha+j\omega)} - \frac{1}{-(\alpha+j\omega)}$$

$$= \frac{1}{\alpha+j\omega}$$

example 8

$FT[\delta(t)]$?

$$\int_{-\infty}^{\infty} \delta(t)\,e^{-j\omega t}\,dt = e^{-j\omega\cdot 0} = 1$$

example 9

$FT[\delta(t-a)]$?

$$\int_{-\infty}^{\infty} \delta(t-a)\,e^{-j\omega t}\,dt \rightarrow e^{-j\omega a}$$

example 10

$$FT[\chi_{(-a,a)}(t)]$$

$$\int_{-a}^{a} e^{-j\omega t} \rightarrow 2 \cdot \frac{e^{-j\omega a} - e^{j\omega a}}{(-j\omega) \cdot 2} = 2 \frac{e^{j\omega a} - e^{-j\omega a}}{2j\omega}$$

$$= 2 \frac{\sin(\omega a)}{\omega}$$

remember: $\sin\theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}$

example 11

$$FT(1)?$$

$$\lim_{\to\infty} \int_{-T}^{T} 1 \cdot e^{-j\omega t} dt = \lim_{T\to\infty} \left(\frac{e^{-j\omega t}}{-j\omega} \Big|_{-T}^{T} \right)$$

$$\hookrightarrow \frac{e^{-j\omega T} - e^{j\omega T}}{-j\omega}$$

$$\lim_{T\to\infty} 2 \cdot \left(\frac{e^{-j\omega T} - e^{j\omega T}}{(-j\omega) \cdot 2} \right) = \lim_{T\to\infty} 2 \cdot \frac{\sin(\omega T)}{\omega} = 2\pi\delta(\omega)$$

because:

$T = \frac{1}{\varepsilon}$:

$$2 \cdot \lim_{\varepsilon\to 0} \frac{\sin(\frac{\omega}{\varepsilon})}{\omega}$$

$\jmath = i$ (aynı şey)

example 12: $FT[e^{\jmath\alpha t}] \quad \alpha \in R$

$$\int_{-\infty}^{\infty} e^{\jmath\alpha t} e^{+\jmath\omega t}\, d\jmath = \int_{-\infty}^{\infty} e^{\jmath(\alpha-\omega)t}\, dt$$

(problematic on a± 0. use p limit)

like in prev. example

$$\hookrightarrow \lim_{T\to\infty} \int_{-T}^{T} e^{\jmath(\alpha-\omega)t}\, dt$$

$$= \lim_{T\to\infty} \left(\frac{e^{\jmath(\alpha-\omega)t}}{\alpha-\omega} \Big|_{-T}^{T} \right) = \lim_{T\to\infty} \frac{e^{\jmath(\alpha-\omega)T} - e^{-\jmath(\alpha-\omega)T}}{(\alpha-\omega)2}$$

$$= 2\pi\,\delta(\alpha-\omega) = 2\pi\,\delta(\omega-\alpha)$$

example 13: $FT[\sin(\alpha t)]$?

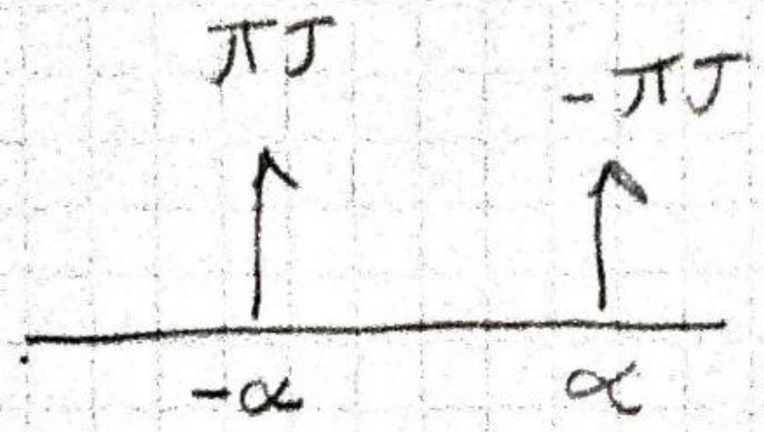
$$= FT\left[\frac{e^{\jmath\alpha t} - e^{-\jmath\alpha t}}{2\jmath}\right] = \frac{FT(e^{\jmath\alpha t}) - FT(e^{-\jmath\alpha t})}{2\jmath}$$

$$\frac{2\pi\,\delta(\omega-\alpha) - 2\pi\,\delta(\omega+\alpha)}{2\jmath}$$

$\frac{1}{\jmath} = -\jmath$

$$= \pi\jmath\left[\delta(\omega+\alpha) - \delta(\omega-\alpha)\right]$$

plot on next page



example 14:

$$FT[\cos(\alpha t)]$$

$$= FT\left[\frac{e^{J\alpha t} + e^{-J\alpha t}}{2}\right] = \frac{2\pi\delta(\omega-\alpha) + \delta(\omega+\alpha)\,2\pi}{2}$$

$$= \pi[\delta(\omega-\alpha) + \delta(\omega+\alpha)]$$

example 15:

$$FT[u(t)] ?$$

$$\int_{-\infty}^{\infty} u(t)\, e^{-J\omega t}\, dt \rightarrow \int_{0}^{\infty} e^{-J\omega t}\, dt = \frac{e^{-J\omega\infty} - e^{-J\omega\cdot 0}}{-J\omega}$$

$$= \frac{1}{J\omega} \; ??$$

★ olmadı...

$$\lim_{T\to\infty} \int_{0}^{T} e^{-J\omega t}\, dt = \frac{1}{J\omega} - \lim_{T\to\infty} \frac{e^{-J\omega t}}{J\omega} \rightarrow$$

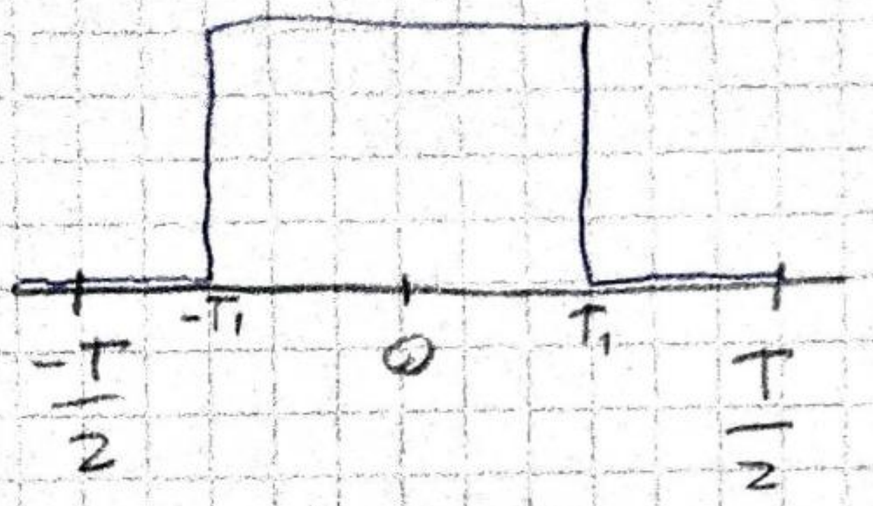
remember:
$\cos(-\omega t) + J\sin(-\omega t)$

$$= \frac{1}{J\omega} - \lim_{T\to\infty}\left[\frac{\cos(\omega T)}{J\omega} + \frac{J\sin(\omega T)}{J\omega}\right]_0$$

(we have learned it)

$$= \frac{1}{j\omega} + \lim_{T\to\infty} \frac{\sin(\omega T)}{\omega} = \frac{1}{j\omega} + \pi\,\delta(\omega)$$

example 16



Find FT!

Answer:

$$\rightarrow a_k = \frac{1}{T}\int_{-T/2}^{T/2} \underbrace{x_{(-T_1,T_1)}(t)}\, e^{-\frac{2\pi}{T}kt\,j}\, dt \longrightarrow \frac{2\pi}{T} = \omega_0$$

(Fourier Series)

$$a = \frac{1}{T}\int_{-T_1}^{T_1} e^{-k\omega_0 t\,j}\, dt = \frac{1}{T}\,\frac{e^{-k\omega_0 t\,j}}{-k\omega_0 j}\Bigg|_{-T_1}^{T_1} = \frac{2}{T}\,\frac{e^{-k\omega_0 T_1 j} - e^{jk\omega_0 T_1}}{2(-k\omega_0 j)}$$

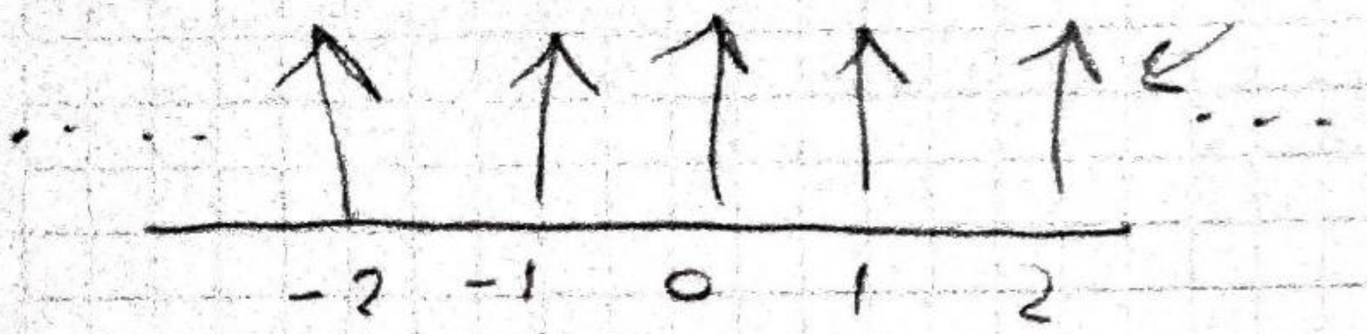
$$= \frac{2}{T}\,\frac{\sin(k\omega_0 T_1)}{k\omega_0} = \frac{2\sin(k\omega_0 T_1)}{k\omega_0 T_0}$$

$$\therefore FT(\text{periodic}) = 2\pi \sum_{k=-\infty}^{\infty} a_k\,\delta(\omega - k\omega_0)$$

$$= 2\pi \sum_{k=-\infty}^{\infty} \frac{2\sin(k\omega_0 T_1)}{k\omega_0 T_0}\,\delta(\omega - k\omega_0)$$

(Plot on next page)

$$2\pi \frac{2\sin(2\omega_0 T_1)}{2\omega_0 T_0}$$

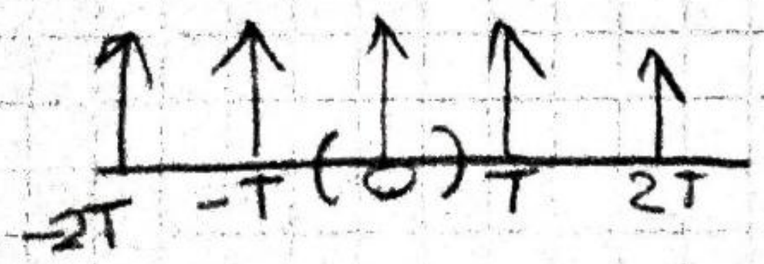


example 17

$$FT\left[\sum_{k=-\infty}^{\infty} \delta(t-Tk)\right] ?$$

Periodic

Answer:



$$a_k = \frac{1}{T}\int_{-T/2}^{T/2} \delta(t)\, e^{-Tk\omega_0 t}\, dt$$

$$= \frac{1}{T}$$

$$FT = 2\pi \sum_{k=-\infty}^{\infty} \frac{1}{T}\,\delta\left(\omega - \underbrace{\frac{2\pi}{T}}_{\omega_0} k\right)$$

Remember:

$$\boxed{X(\omega) = \sum_{k=-\infty}^{\infty} 2\pi a_k\, \delta(\omega - k\omega_0)}$$

Properties of Fourier Transforms (continuous)

1) Linearity

$$f(t) \xrightarrow{\text{fourier}} F(\omega)$$
$$g(t) \longrightarrow G(\omega)$$

$$FT[\alpha f(t) + \beta g(t)] = \alpha F(\omega) + \beta G(\omega)$$

2.) Time Shift

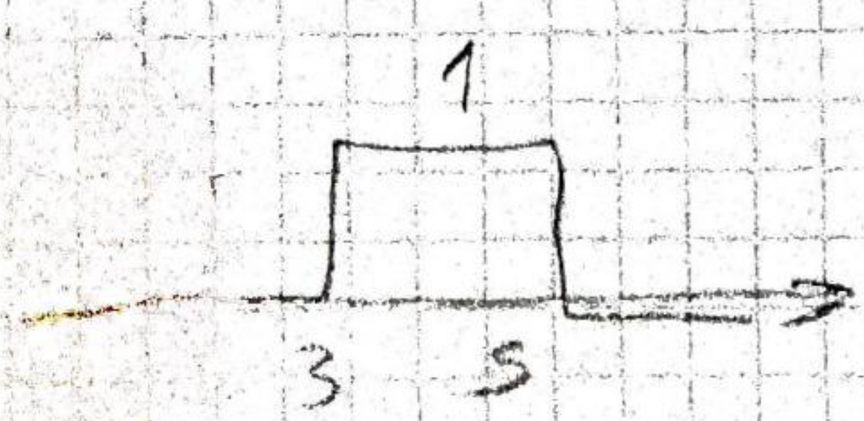
$$x(t-a) = e^{-j\omega a} X(\omega)$$

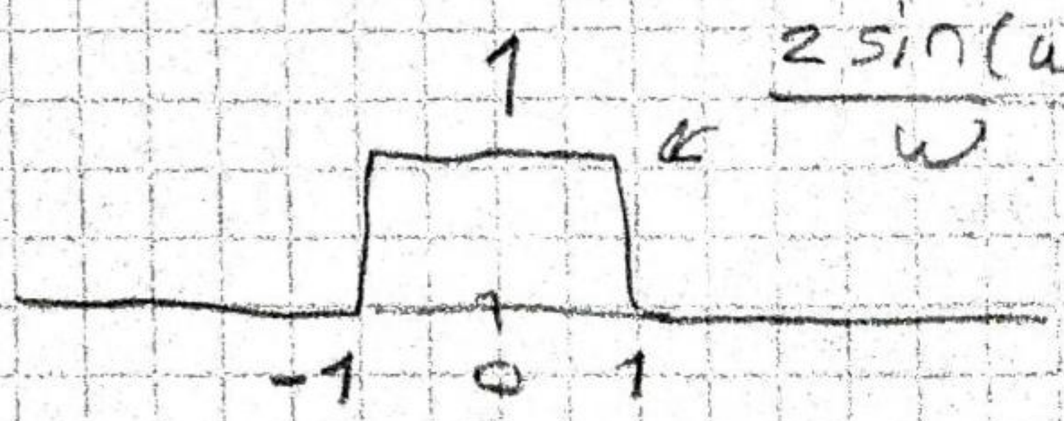
ex:

$$e^{-2(t-5)} u(t-5) \xrightarrow{FT} e^{-j\omega 5} \frac{1}{j\omega+2}$$

$$e^{-2t} u(t) \longrightarrow \frac{1}{j\omega+2}$$

ex 2:





$$e^{-j\omega 4} \frac{2\sin(\omega)}{\omega}$$

3) $e^{Jat} x(t) \xrightarrow{FT} X(w-a)$

4) $x(-t) \xrightarrow{Ft} X(-w)$

5) $x(t) \longrightarrow X(w)$

(scaling)

$x(at) \longrightarrow \frac{1}{|a|} X\left(\frac{w}{a}\right)$

6) $x(t) \longrightarrow X(w)$ $\quad y(t) \longrightarrow Y(w)$

$x(t) * y(t) \longrightarrow X(w)\,Y(w)$

(convolution)

7) multiplication in time is convolution in frequency (reverse of above)

$x(t)\,y(t) \longrightarrow \frac{1}{2\pi} X(w) * Y(w)$

8) $\frac{d}{dt} x(t) \longrightarrow Jw\,X(w)$

9) $\int_{-\infty}^{t} x(s)\,ds \longrightarrow \frac{1}{Jw} X(w) + \pi X(0)\,\delta(w)$